Income and Wealth Inequality in America, 1949–2016

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This paper introduces a new long-run data set based on archival data from historical waves of the Survey of Consumer Finances. Studying the joint distribution of household income and wealth, we expose the central importance of portfolio composition and asset prices for wealth dynamics in postwar America. Asset prices shift the wealth distribution because of systematic differences in household portfolios along the wealth distribution. Middle-class portfolios are dominated by housing, while rich households predominantly own business equity. Differential changes in equity and house prices shaped wealth dynamics in postwar America and decoupled the income and wealth distribution over extended periods.

I. Introduction

We live in unequal times. The causes and consequences of widening disparities in income and wealth have become a defining debate of our age.

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Recent studies have made major inroads into documenting trends in either income or wealth inequality in the United States (Piketty and Saez 2003; Kopczuk, Saez, and Song 2010; Saez and Zucman 2016), but we still know little about how the joint distributions of income and wealth evolved over the long run. This paper fills this gap.

The backbone of this study is a newly compiled data set that builds on household-level information and spans the entire US population over seven decades of postwar American history. We unearthed historical waves of the Survey of Consumer Finances (SCF) that were conducted by the Economic Behavior Program of the Survey Research Center at the University of Michigan from 1947 to 1977. In extensive data work, we linked the historical survey data to the modern SCFs that the Federal Reserve redesigned in 1983.[1] We call this new resource for inequality research the SCF+.

The SCF+ complements existing data sets for long-run inequality research that are based on income tax and social security records but also goes beyond them in a number of important ways. Importantly, the SCF+ is the first data set that makes it possible to study the joint distributions of income and wealth over the long run. As a historical version of the SCF, it contains the same comprehensive income and balance sheet information as the modern SCFs. This means that we do not have to combine data from different sources or capitalize income tax data to generate wealth holdings. Moreover, the SCF+ contains granular demographic information that can be used to study dimensions of inequality —such as long-run trends in racial inequality— that so far have been out of reach for research.

Our analysis speaks to the quest to generate realistic wealth dynamics in dynamic quantitative models (Gabaix et al. 2016; Fella and De Nardi

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[1] A few studies, such as Malmendier and Nagel (2011) and Herkenhoff (2013), exploited parts of these data to address specific questions, but no study has attempted to harmonize modern and historical data in a consistent way. Note that we leave the post-1983 modern SCF unchanged. Its value for studying distributional trends has been demonstrated in recent contributions by Bricker et al. (2016) and Wolff (2017).

2017; Hubmer, Krusell, and Smith 2017; Benhabib and Bisin 2018). A key finding of our paper is that a channel that has attracted little scrutiny so far has played a central role in the evolution of wealth inequality in postwar America: asset price changes induce large shifts in the wealth distribution. This is because the composition and leverage of household portfolios differ systematically along the wealth distribution. While the portfolios of rich households are dominated by corporate and noncorporate equity, the portfolio of a typical middle-class household is highly concentrated in residential real estate and, at the same time, highly leveraged. These portfolio differences are persistent over time. We document this stylized fact and expose its consequences for the dynamics of the wealth distribution.

An important upshot is that the top and the middle of the distribution are affected differentially by changes in equity and house prices. Housing booms lead to substantial wealth gains for leveraged middle-class households and tend to decrease wealth inequality, all else equal. Stock market booms primarily boost the wealth of households at the top of the wealth distribution, as their portfolios are dominated by listed and unlisted business equity. Portfolio heterogeneity thus gives rise to a race between the housing market and the stock market in shaping the wealth distribution. We show that over extended periods in postwar American history, such portfolio valuation effects have been predominant drivers of shifts in the distribution of wealth.

A second consequence of portfolio heterogeneity is that asset price movements can introduce a wedge between the evolution of the income and wealth distribution. For instance, rising asset prices can mitigate the effects that low income growth and declining savings rates have on wealth accumulation. Looking at income and wealth growth of different parts of the wealth distribution, we find that such a divergence played a prominent role in the four decades before the financial crisis. The middle class (50th–90th percentiles) rapidly lost ground to the top 10% with respect to income but, by and large, maintained its wealth share thanks to substantial gains in housing wealth. The SCF+ data show that incomes of the top 10% grew 80% more than incomes of middle-class households (50th–90th percentiles) and 120% more than incomes in the bottom 50% of households. In line with previous research, the SCF+ data thus confirm a strong trend toward growing income concentration at the top (Piketty and Saez 2003; Kopczuk, Saez, and Song 2010). However, when it comes to wealth, the picture is different. For the bottom 50% of the wealth distribution, wealth grew 100% in excess of income between 1971 and 2007. A particularly pronounced difference using consumer price index (CPI) inflation that leads to zero income growth and a doubling of wealth. For the middle class and for the top 10%, wealth grew at approximately the same rate despite diverging income paths. As a

result, wealth-to-income ratios increased most strongly for the bottom 90% of the wealth distribution. That the SCF+ data reach back to the 1950s and 1960s—that is, before the income distribution started to widen substantially—makes it possible to expose these divergent trends.

Importantly, price effects account for a major part of the wealth gains of the middle class and the lower middle class. We estimate that between 1971 and 2007, wealth of the bottom 50% grew almost entirely because of price effects—essentially a doubling of wealth compared with household income without any (active) saving. Price-related wealth growth is high for the bottom 50% despite below-average homeownership rates because virtually all existing wealth of this group is invested in highly leveraged housing wealth. Even in the middle and at the top of the distribution, asset price–induced gains accounted for close to half of total wealth growth over the 1971–2007 period, comparable to the contribution of savings flows. From a political economy perspective, it is conceivable that the strong wealth gains for the middle and lower middle class helped to dispel discontent about stagnant incomes. They may also help to explain the disconnect between trends in income and consumption inequality that have been the subject of some debate (Attanasio and Pistaferri 2016). When house prices collapsed in the 2008 crisis, the same leveraged portfolio position of the middle class brought about substantial wealth losses, while the quick rebound in stock markets boosted wealth at the top. Relative price changes between houses and equities after 2007 have produced the largest spike in wealth inequality in postwar American history. Surging postcrisis wealth inequality might in turn have contributed to the perception of sharply rising inequality in recent years.

Thanks to its demographic detail, we can also exploit the SCF+ to shed new light on the long-run evolution of racial inequalities. The SCF+ covers the entire postwar history of racial inequality and spans the pre– and post–civil rights eras. With information on income and wealth at the household level, not only do we complement recent studies of the long-run evolution of racial wage inequality (Bayer and Charles 2017) but also we add new dimensions. Most importantly, the SCF+ data offer a window on long-run trends in racial wealth inequality that have so far remained uncharted territory. We expose persistent and, in some respects, growing inequalities between black and white Americans. Income disparities today are as big as they were in the pre–civil rights era. In 2016, black household income is still only half of the income of white households. The racial wealth gap is even wider and is still as large as it was in the 1950s and 1960s. The median black household persistently has less than 15% of the wealth of the median white household. We also find that the financial crisis has hit black households particularly hard and has undone the little progress that had been made in reducing the

racial wealth gap during the 2000s (Wolff 2017). The overall summary is bleak. The typical black household remains poorer than 80% of white households.

Related literature.—Research on inequality has become a highly active field, and our paper speaks to a large literature. Analytically, the paper is most closely related to recent contributions emphasizing the importance of heterogeneity in returns on wealth for the wealth distribution. On the empirical side, this literature has mainly worked with European data, while our paper addresses the issues with long-run micro data for the United States. Bach, Calvet, and Sodini (2016) study administrative Swedish data. With regard to heterogeneity in returns along the wealth distribution, Fagereng et al. (2016) use administrative Norwegian tax data and document substantial heterogeneity in wealth returns and intergenerational persistence. For France, Garbinti, Goupille-Lebret, and Piketty (2017) analyze the long-run distribution of wealth as well as the role of return and savings rate differentials. In the American context, Wolff (2016) demonstrates the sensitivity of middle-class wealth to the house price collapse in the Great Recession, and his earlier research (Wolff 2002) is closely related as it discusses the sensitivity of the US wealth distribution to asset price changes. In the policy debate, the role of asset prices for the wealth distribution has also been discussed, for example, by Yellen (2014). Moreover, Kuhn and Ríos-Rull (2016) argue that housing wealth plays an important role for the wealth distribution.

With respect to data production and the emphasis on long-run trends, our paper complements the pioneering work of Piketty and Saez (2003), Kopczuk and Saez (2004), and Saez and Zucman (2016) as well as the work of Kopczuk, Saez, and Song (2010). Our paper also speaks to the more recent contribution of Piketty, Saez, and Zucman (2017), who combined micro data from tax records and household survey data to derive the distribution of income reported in the national accounts. Saez and Zucman (2016) estimate the wealth distribution by capitalizing income flows from administrative data. This approach is advantageous for households at the top of the distribution that hold a significant part of their wealth in assets that generate taxable income flows. Yet many assets in middle-class portfolios do not generate taxable income flows—housing being a prime example. The SCF+ provides long-run data on all sources of income (including capital and nontaxable income) as well as the entire household balance sheet with all assets (including residential real estate) and liabilities (including mortgage debt). Playing to the strength of our data, our paper focuses on the bottom 90% of households, not on changes in inequality at the very top. We also connect our paper to the recent paper by Bricker et al. (2016) that demonstrates the potential of the modern SCFs to study distributional trends even at the top, and we

discuss the differences between the more advanced modern SCF and the historical SCF waves.[2]

Theoretical work modeling the dynamics of wealth inequality has grown quickly. A common thread is that models based on labor income risk alone typically produce too little wealth concentration and cannot account for substantial shifts in wealth inequality that occur over short time horizons. Our paper speaks to recent work by Gabaix et al. (2016), Benhabib, Bisin, and Luo (2017), and Benhabib and Bisin (2018), who discuss the importance of heterogeneous returns for the wealth distribution and its changes over time. In another recent paper, Hubmer, Krusell, and Smith (2017) use variants of incomplete markets models to quantify the contribution of different drivers for rising wealth inequality and point to return differences and portfolio differences as a neglected line of research. Our findings support the emphasis on asset returns.[3] Glover et al. (2017) quantify the welfare effects of wealth changes resulting from portfolio differences and asset price changes during the Great Recession. Fella and De Nardi (2017) survey the existing literature and discuss different models from the canonical incomplete market model to models with intergenerational transmission of financial and human capital, rate of return risk on financial investments, and more sophisticated earnings dynamics.

The paper is divided into three parts. Section II documents the extensive data work that we have undertaken over the past years to construct the SCF+ and what we did to align the historical and modern SCF data. Section III then exploits the new data and presents stylized facts for long-run trends in income and wealth inequality, including racial inequalities, that emerge from the SCF+. Section IV studies the joint distributions of income and wealth and exposes the central importance of asset price changes for the dynamics of the wealth distribution in postwar America. Section V concludes.

II. Constructing the SCF+

The SCF is a key resource for research on household finances in the United States. It is a triennial survey, and the post-1983 data are available

[2] Work in labor economics often relies on data from the Current Population Survey (CPS). Examples are Gottschalk and Danziger (2005) and Burkhauser, Feng, and Jenkins (2009). Most relevant for our work is Burkhauser et al. (2012), who show that trends in income inequality derived from the CPS are similar to the inequality series based on tax data in Piketty and Saez (2003). They also provide a detailed discussion of the conceptual differences in measuring income in the tax and CPS data.

[3] See also Castaneda, Díaz-Giménez, and Ríos-Rull (2003) for a benchmark model of cross-sectional income and wealth inequality and Kaymak and Poschke (2016) for another recent attempt to explain time trends. Heathcote, Perri, and Violante (2010) and Díaz-Giménez, Glover, and Ríos-Rull (2011) study the evolution of income inequality in recent decades.

on the website of the Board of Governors of the Federal Reserve System.[4] Yet the first consumer finance surveys were conducted as far back as 1947. The early SCF waves were directed by the Economic Behavior Program of the Survey Research Center of the Institute for Social Research at the University of Michigan. The surveys were taken annually between 1947 and 1971 and then again in 1977. The raw data are kept at the Inter-University Consortium for Political and Social Research at the Institute for Social Research in Ann Arbor, Michigan.

For this paper, we linked the archival survey data to the post-1983 SCF. To do this, we harmonized and reweighted the historical data to make them as compatible as possible with the modern SCF. Note that we do not adjust the post-1983 SCF data. On the contrary, we take the advanced survey design of the modern SCF as the benchmark and adjust the historical surveys so that they come as close as possible to this benchmark. We discuss in detail below and in appendix B (apps. A–E are available online) how we proceeded and how consistent the historical and modern data are, especially when it comes to the top of the distribution. The combined data set adds four decades of household-level micro data, effectively doubling the time span covered by the SCF. As a new resource for long-run research on household finances, we refer to this historically extended version of the SCF as the SCF+.

The SCF+ complements the data sets for long-run trends in the distribution of income and wealth in the United States that Piketty and Saez (2003), Kopczuk and Saez (2004), and Saez and Zucman (2016) have compiled using administrative tax data. Other researchers have used the 1962 Survey of Financial Characteristics of Consumers (SFCC) that provides a snapshot of the financial conditions of US households in 1962 (Wolff 2017).[5] But so far, the tax data constitute the only data covering the entire postwar period on a continuous basis. The SCF+ provides an opportunity to corroborate and improve our understanding of postwar trends in the distribution of income and wealth.

For future researchers, it is important to have a good understanding of the relative strengths and weaknesses of the SCF+ for inequality research. A key advantage of the long-run tax data is their compulsory collection process resulting in near-universal coverage at the top of the distribution, whereas survey data have to cope with nonresponse of rich households. This being said, the tests carried out in a recent paper by Bricker et al. (2016) show that the modern SCF with its combined administrative and survey data methodology also captures households at the very top of the distribution.

[4] https://www.federalreserve.gov/econres/scfindex.htm. See Bricker et al. (2017) for results from the 2017 SCF data and for general information on the SCF data and sampling.

[5] For the construction of the SCF+, we have set the distributional information from the 1962 SCF against the SFCC data and generally found the differences to be small.

The strengths of the administrative data in terms of accuracy and coverage at the top of the distribution also have to be weighed against the attractive properties of survey data in other respects. Most importantly, the survey data contain direct measurements of assets and debt plus the information to stratify households by demographic characteristics. The survey data also cover people who do not file taxes, and the unit of analysis is the household, not the tax unit. This structure is in line with economic models in which the household is the relevant unit for risk and resource sharing.[6]

Moreover, specific challenges arise when income tax data are used to construct wealth estimates. The capitalization method of Saez and Zucman (2016) relies on observable income tax flows that are capitalized to allocate aggregate wealth positions in the cross section. While ingenious as an approach, some gaps remain because a substantial part of wealth does not generate taxable income flows and has to be imputed (often on the basis of survey data). The key asset here is owner-occupied housing as well as its corresponding liability, mortgage debt. Pension assets also do not generate taxable income flows, and unrealized capital gains do not show up on tax returns until they are realized.

In the estimates of Saez and Zucman (2016), about 90% of the total wealth outside the top 10% has to be imputed. And even for the top 10%, the share of imputed wealth stands at 40%. Saez and Zucman (2016) correctly stress that the exact distribution of these assets is of minor importance for the very top of the wealth distribution. Yet for researchers interested in long-run distributional changes outside the very top, these are binding constraints that the SCF+ overcomes. The capitalization method also has to derive returns for individual asset classes from a combination of capital income from tax data and aggregate estimates from the financial accounts. Kopczuk (2015) provides an illustration of how this method can lead to an upward bias of wealth concentration during low interest rate periods, and the recent paper by Bricker, Henriques, and Hansen (2018) quantifies this bias and discusses in detail other conceptual differences between survey estimates and estimates based on tax data.

A. Variables in the SCF+

The variables covered in the historical surveys of the SCF+ correspond to those in the contemporary SCF, but the exact wording of the questions can differ from survey to survey. Some variables are not continuously

[6] In 2012, there were about one-third more tax units (160.7 million) than households (121.1 million) in the United States. Bricker et al. (2016) argue that relying on tax units could lead to higher measured income concentration toward the top of the distribution.

covered, so we have to impute values in some years. We explain the imputation procedure in the following section. Our analysis focuses on the four variables that are of particular importance for household finances: income, assets, debt, and wealth. In the analysis, we use all data and abstain from any sample selection. We adjust all data for inflation using the CPI and report results in 2016 dollars.[7] Table 1 provides a general overview of variables and years when imputation is used. Appendix A.1 contains additional information.

1. Income

We construct total income as the sum of wages and salaries, income from professional practice and self-employment, rental income, interest, dividends, transfer payments, as well as business and farm income. Note that we do not include imputed rental income of homeowners in the baseline, but we provide additional results in appendix D.2.

2. Assets

The historical SCF waves contain detailed information on household assets. We group assets into the following categories: liquid assets, housing, bonds, stocks and business equity, mutual funds, the cash value of life insurance, defined-contribution retirement plans, other real estate, and cars.[8] Liquid assets comprise the sum of checking, savings, call/money market accounts, and certificates of deposits. By contrast, Social Security as a key asset for most families is not measured as part of household wealth. The wealth concept used here hence follows the literature by focusing on marketable household wealth. A more detailed discussion of the importance of Social Security for household wealth and its distribution can be found in Bricker et al. (2016).

[7] We use CPI data from the Macrohistory Database (Jordà, Schularick, and Taylor 2017). The series combines the CPI research series using current methods (CPI-U-RS; 1978–2016) from the Bureau of Labor Statistics and the CPI for all urban consumers (CPI-U) for 1948–77. The CPI shows higher inflation rates relative to the personal consumption expenditure index (PCE), as discussed by Furth (2017). Comparisons of relative income and wealth trends between groups are unaffected by the choice of the deflator, but caution is warranted for absolute statements about income and wealth growth. We provide a sensitivity analysis using the PCE in app. D.4.

[8] Data on defined-contribution retirement plans are available only from 1983 onward. However, according to the financial accounts of the United States (FA), this variable makes up a small part of household wealth before the 1980s, so missing information before 1983 is unlikely to change the picture meaningfully. Up to 1970, defined-contribution plans correspond to less than 1% of average household wealth.

TABLE 1
Data Availability across Survey Years

Survey Year	Income			Financial Assets			Nonfinancial Assets			Debt			
	Total	Labor	Labor and Business	Liquid Assets	Bonds	Equity	Housing	Other Real Estate	Business	Total	Housing	Other Real Estate	Nonhousing
1949	O	O	O	O	O	O	O	I	I	O	O	O	O
1950	O	O	O	O	O	O	O	O	O	O	O	O	O
1951	O	O	O	O	O	I	O	I	I	O	O	O	O
1953	O	O	O	O	O	O	O	O	O	O	O	O	O
1954	O	O	O	O	O	I	O	I	I	O	O	O	O
1955	O	O	O	O	O	O	O	I	I	O	O	O	O
1956	O	O	O	O	O	I	O	I	I	O	O	I	O
1957	O	O	O	O	O	I	O	I	I	O	O	O	O
1958	O	O	O	O	O	I	O	I	I	O	O	O	O
1959	O	O	O	O	O	I	O	I	I	O	O	O	O
1960	O	I	O	O	O	O	O	O	O	O	O	I	O
1962	O	I	O	O	O	O	O	O	O	O	O	I	O
1963	O	I	O	O	O	O	O	O	O	O	O	I	O
1965	O	I	O	O	O	I	O	I	I	O	O	I	O
1967	O	O	O	O	I	O	O	I	I	O	O	I	O
1968	O	O	O	O	I	O	O	O	I	O	O	O	O
1969	O	O	O	O	I	O	O	O	I	O	O	O	O
1970	O	O	O	O	O	O	O	O	O	O	O	O	O
1971	O	O	I	O	I	I	O	I	I	O	O	I	O
1977	O	O	I	O	O	O	O	O	I	O	O	O	O

Note.—"O" indicates that original observations from that survey year are used, and "I" indicates that the variable has been imputed using data from other survey years. In some years, totals are available, but components are not separately reported and had to be imputed. If the total is constructed as the sum of components, then totals are marked as imputed if any component is imputed. Equity includes stocks and mutual funds.

3. Debt

Total debt consists of housing and nonhousing debt. Housing debt is calculated as the sum of debt on owner-occupied homes, home equity loans, and home equity lines of credit. For 1977, only the origination value (instead of the current value) of mortgages is available. Using information on the year the mortgage was taken out, remaining maturity, and an estimated annual interest rate, we create a proxy for debt on homes for 1977. All debt other than housing debt refers to and includes car loans, education loans, and other consumer loans.

4. Wealth

We construct wealth as the consolidated value of the household balance sheet by subtracting debt from assets. Wealth constitutes households' net worth.

B. Weights and Imputations

The SCF is designed to be representative of the US population. As Bricker et al. (2016) discuss, the modern SCF applies a sophisticated dual-frame sampling scheme to oversample wealthy households, combining administrative and survey data. The historical surveys did not oversample households at the top. In this section and in appendix B, we outline how we dealt with the issue and discuss the implications for the representativeness of the SCF+. In addition to the adequate coverage of wealthy households, we also need to ensure representative coverage of demographic characteristics, such as race, age, and education.

1. Oversampling of Wealthy Households

Since its redesign in 1983, the SCF consists of two samples. The first sample is drawn using area probability sampling of the entire US population based on census information. In addition, a second so-called list sample is drawn based on tax information.[9] For both samples, survey weights are constructed separately. In the list sample, survey weights have to be disproportionally adjusted for nonresponse. The weight of each household corresponds to the number of similar households in the population. In a final step, both samples are combined and survey weights are adjusted so that the combined sample is representative of the US population (Kennickell and Woodburn 1999).

[9] The methodology has evolved over time and uses a combination of income capitalization with income from several tax years and regression evidence based on existing surveys. See Bricker et al. (2017) and Kennickell (2017) for details and further references.

Before 1983, the historical SCF data are not supplemented by a list sample. As a consequence, the challenge of adequately representing households at the very top is likely to be more pervasive (Sabelhaus et al. 2015). Missing households at the top could potentially lead to an underrepresentation and bias historical inequality measures downward.

For the construction of the SCF+, we use information from the 1983 list sample to adjust the coverage of rich households in the pre-1983 data. In a first step, we determine the proportion of households in the 1983 list sample relative to all households. Their share corresponds to approximately 2%. In a second step, we determine where the households from the list sample are located in the income and wealth distribution in 1983. We find that most observations are among the top 5% of the income and wealth distribution. Using this information, we adjust survey weights in all survey years before 1983 in two steps. First, for each year we extract all observations that are simultaneously in the top 5% of the income and wealth distribution. Second, we increase the weighting of these households in such a way that we effectively add 2% of wealthy households to the sample and adjust the remaining weights accordingly. This approach is similar in spirit to Bricker, Henriques, and Hansen (2018), who adjust SCF weights inversely proportional to the overlap of the SCF sample with the Forbes list.

The list sample of the modern SCF is concentrated in the top 1% of the wealth distribution, and great effort goes into identifying these households as Bricker, Henriques, and Hansen (2018) discuss. A potential concern with our adjustment of the historical data is that we can only increase the weight of households that are sampled. This could be problematic if nonresponse rates have changed over time or if the older surveys failed to identify and contact wealthy households in sufficient numbers. One way to get a better sense of how pervasive these issues are is to compare the 1983 data with the 1962 SFCC. The SFCC was the only historical survey that also used a dual-frame sampling scheme similar to the 1983 list sample on the basis of income tax records.

Table 2 compares the nonresponse patterns at the top of the income and wealth distribution from these two surveys. Importantly, we find little

TABLE 2
Share of Respondents (%) from List Sample at Top of Distribution

	Income		Wealth	
	Top 10%	Top 5%	Top 10%	Top 5%
SFCC 1962	21	35	20	28
SCF 1983	17	34	17	32

Note.—Income shows the shares of households in the top 10% and top 5% of the income distribution in the 1962 SFCC and the 1983 SCF data from the list sample. Wealth shows the corresponding shares for the top 10% and top 5% of the wealth distribution.

evidence for a pronounced time trend in nonresponse of wealthy households. For the modern SCF surveys, the reported response rates for the list sample also do not point to any trends in nonresponse rates for the list sample over time (see Bricker et al. 2016, 2017). In appendix B, we apply a battery of tests to the historical data that were proposed in a recent paper by Bricker et al. (2016) to examine how well the modern SCFs perform in capturing the top of the distribution relative to the tax data. More precisely, we test how many households in the SCF+ are above the 99th percentile threshold for income and wealth from the tax data and how mean income and wealth above this threshold compare. Although these tests do not signal systematic deviations, the strength of the SCF+ data clearly lies in their comprehensive coverage of the lower ranks of the distribution. A lot of research has already been devoted to small groups at the very top of the distribution, but less is known about long-run distributional trends among the bottom 99% of households. Consequently, in this paper, we will not talk about the top 1% of households but will focus on income and wealth trends of the remaining 99% of American families.

2. Demographic Characteristics

We compare the demographic characteristics in the surveys before 1983 with data from the US CPS. To obtain samples that match the CPS data, we subdivide both the CPS and the SCF+ data into demographic subgroups. Subgroups are determined by age of the household head, college education, and race. In addition to these demographic characteristics, we include homeownership as an additional dimension. We adjust SCF+ weights by minimizing the difference between the share of each subgroup in the SCF+ and the respective share in the CPS. As census data are available only since 1962, we rely on data from the decennial census and linear inter- and extrapolation for the earlier years. Figure 1 shows the shares of 10-year age groups, college households, and black households in the CPS/census (squares) and in the SCF+ with the adjustment of survey weights (circles). Using adjusted weights (diamonds), the distributions of age, education, and race closely match the CPS/census data. We match the homeownership rate equally well after the adjustment (see fig. A.1; figs. A.1, A.2, B.3–B.8, D.9–D.11, E.12–E.14 are available online).

3. Missing Variables

The imputation of missing variables is done by predictive mean matching as described in Schenker and Taylor (1996). This multiple imputation method assigns variable values by finding observations that are closest to the respective missing observations. In line with the post-1983 data,

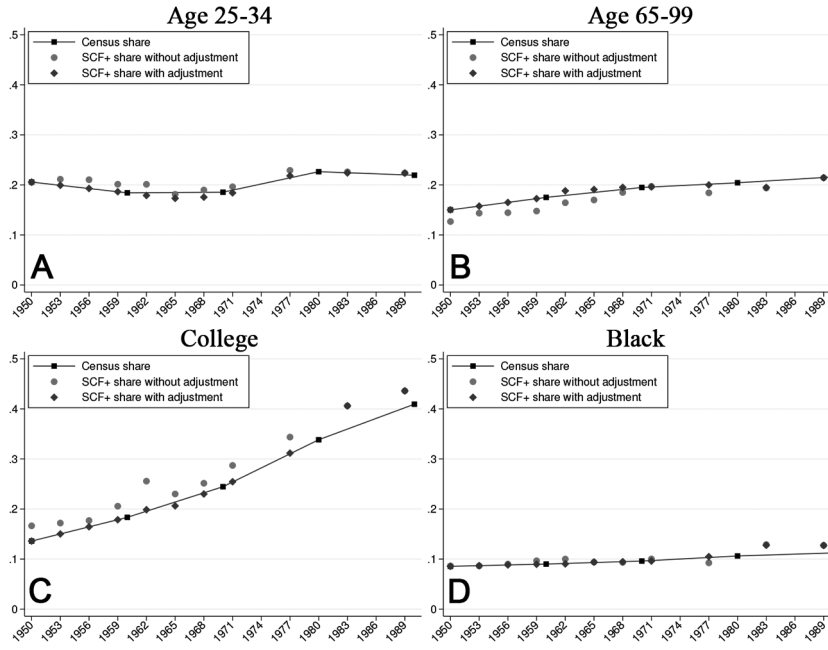


FIG. 1.—Population shares of age groups, college households, and black households. Lines with squares show the population share of the respective demographic group in the CPS/census data. Circles show the population shares of the respective group using the original (unadjusted) survey data. Diamonds show the population shares using the adjusted survey data. A color version of this figure is available online.

we impute five values for each missing observation. We account for a potential undercoverage of business equity before 1983 and follow the method proposed by Saez and Zucman (2016) to adjust the observed holdings in the micro data with information from the financial accounts. A detailed description of these steps is provided in appendix A.1.

Table 1 shows the variables and their coverage as well as the years in which we imputed data.[10] In appendix A.1, we also explain how we impute the value of cars for selected years based on model and purchasing year.

The final SCF+ data set comprises 35 survey years with cross-sectional data, totaling 102,304 household observations. The number of observations varies from a minimum of 1,327 in 1971 to a maximum of 6,482 in

[10] We exclude the survey years 1947, 1948, 1952, 1961, 1964, and 1966 because we lack information on housing, mortgages, and liquid assets. These three wealth components are held by a large fraction of households but can be only poorly inferred from information on other variables. For 1977, we impute income using original data for income intervals.

2010. Table A.1 in the appendix reports the number of observations for all survey years.

C. *Aggregate Trends*

Before looking in detail at the evolution of the income and wealth distributions since World War II, the first step is to benchmark aggregate trends from the SCF+ to the national income and product accounts (NIPA) and the FA. To do so, we have to take into account that even high-quality micro data do not always correspond one-to-one to aggregate data, as measurement concepts differ. We follow Henriques and Hsu (2014) and Dettling et al. (2015) to account for the conceptual differences when constructing income and wealth series. We relegate the details to appendix A.5. For the modern SCF data, Henriques and Hsu (2014) and Dettling et al. (2015) conclude that after accounting for the conceptual differences between micro and macro data, the data align well. They also provide detailed discussions for observed differences. Figure 2 compares indexed time series for average household income and wealth from the SCF+ with the corresponding series constructed from NIPA and FA. Figure 2*A* shows that the trend in income is very similar for SCF+ and NIPA data throughout the 1949–2016 time period. For the base period of 1983–89, the SCF+ matches 87% of income from the NIPA. Looking at wealth, the trends differ only slightly. Before 1995, wealth trends from the SCF+ and FA hardly differ. There appears to be a persistent level shift in the late 1990s that Henriques and Hsu (2014) trace back to differences in business wealth and owner-occupied houses. Looking at different wealth components, we find that financial assets in the SCF+ (fig. 3*A*) increase more strongly in the early 2000s than the corresponding FA values. Henriques and Hsu (2014) attribute most of the difference to the coverage of retirement accounts in the SCF data. Figure 3*B* shows that housing as the most important nonfinancial asset is covered well in the survey data. Debt is the household balance sheet component for which the SCF+ matches the aggregate best, as shown in figure 3*C*. Summing up, the SCF+ matches aggregate trends of NIPA data and FA asset and debt positions. In particular, the SCF+ data and the FA show similar trends for the important categories of housing wealth and mortgage debt.

III. Income and Wealth Inequality in the SCF+

This section presents stylized facts for long-run trends in income and wealth inequality that the SCF+ data bring to light. We begin by documenting the evolution of Gini coefficients for income and wealth as a

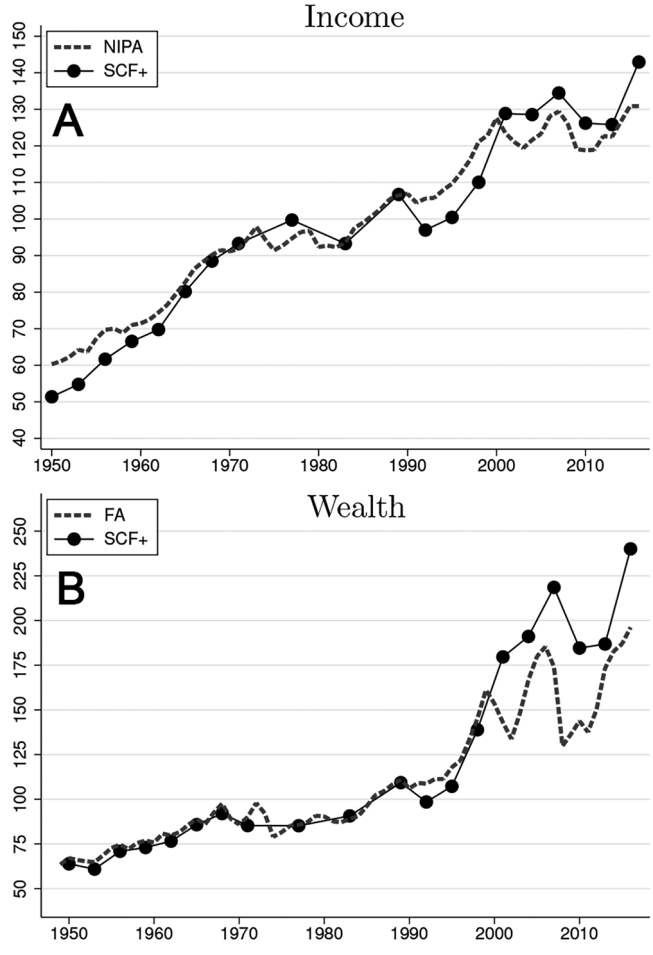


FIG. 2.—Comparison of income and wealth from SCF+, NIPA, and FA data. This figure shows income and wealth data from the SCF+ in comparison to income data from the NIPA and wealth data from the FA. All data have been indexed to the 1983–9 period (=100). Solid lines with circles show SCF+ data, and dashed lines show NIPA and FA data. For the indexing period, SCF+ data correspond to 87% of NIPA income and 90% of FA wealth. A color version of this figure is available online.

comprehensive measure of inequality. We go on and look at the income and wealth inequality trends in different parts of the distribution. For this step, we will rely on the strength of the SCF+ data in covering the bottom 90% of the distribution. We also look at the long-run trends in the share of hand-to-mouth households and use the demographic information in the SCF+ data to analyze the importance of demographic factors in distributional change. Importantly, we present evidence on

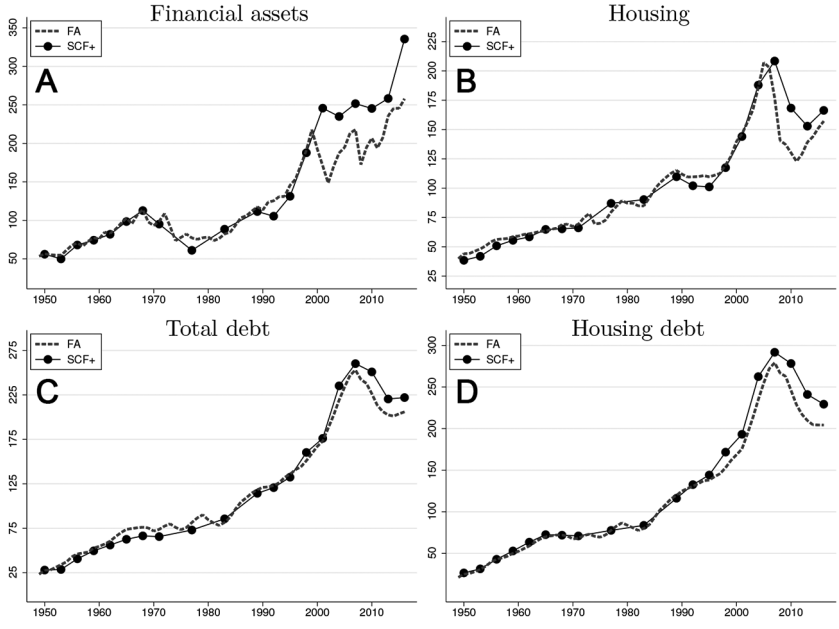


FIG. 3.—Comparison of asset and debt components from SCF+ and FA data. This figure shows asset and debt components of household balance sheets from SCF+ and FA data. All data have been indexed to the 1983–9 period (=100). Solid lines with circles show SCF+ data, and dashed lines show FA data. For the indexed period, SCF+ data correspond to 68% of financial assets, 98% of housing, 73% of total debt, and 75% of housing debt from the FA data. A color version of this figure is available online.

long-run trends in inequalities in income and wealth between black and white households.

A. *Gini Coefficients*

The Gini coefficient is a comprehensive summary measure of inequality along the entire distribution. Table 3 reports Gini coefficients for income and wealth at selected points in time. The first row reports the Gini coefficient for all households; the other rows focus on the bottom 99% and the bottom 90%, respectively.[11]

The Gini coefficients show that income and wealth inequality has increased not only across the entire population (across all households)

[11] We report the full time series in table E.5 in the appendix. We include negative-wealth households in the calculation. Figure D.9 shows that time trends are very similar when we restrict the analysis to positive income and wealth households.

TABLE 3
GINI COEFFICIENT (×100) FOR INCOME AND WEALTH

	1950	1971	1989	2007	2016
Income distribution:					
All households	44	43	53	55	58
Bottom 99%	39	39	46	47	49
Bottom 90%	32	33	39	38	39
Wealth distribution:					
All households	83	79	79	82	86
Bottom 99%	75	74	72	74	79
Bottom 90%	64	62	62	63	70

but also among the bottom 99% and bottom 90% of households. The overall income Gini has risen from its postwar low of 0.43 in 1971 to 0.58 in 2016 (fig. 4*A*). Unsurprisingly, there is a substantial drop in inequality once we look at the bottom 99% of the distribution, but the increase in the Gini coefficient is still substantial. Also, within the bottom 90%, income inequality has widened, yet this has mainly occurred between 1971 and 1989. In section III.C, we explore in detail changes within the bottom 90% over time.[12]

Turning to wealth, it is well known that wealth is considerably more unequally distributed than income. The wealth Gini has fluctuated around 0.8 for most of the postwar period and did not change much, if at all, between 1950 and 2007 (fig. 4*B*). By 2007, it stood at 0.82 and was only marginally higher than in both 1950 and 1971. The marked decline in the wealth Gini between 1971 and 1977 stands out. We will trace this decline back to asset price shifts in section IV.C. A substantial increase in the Gini coefficient occurred between 2007 and 2016, and the wealth Gini reached its postwar peak in 2016.

The confidence bands in figure 4 also show that Gini coefficients for both income and wealth are tightly estimated, although the confidence bands are somewhat wider in the historical data.[13] The observed long-run trends are clearly statistically significant. America is considerably more unequal today than it was in the 1970s with respect to both income and wealth.

[12] Our baseline income does not include rental income of owner-occupiers. As a sensitivity check, we imputed this rental income using historical rental yields from Jordà et al. (2019) in app. D.2. We find the Gini coefficient for income after imputing rents to be slightly lower.

[13] All confidence bands are computed using 999 replicate sample weights. Replicate weights are provided for the modern SCF surveys after 1983. For the historical surveys, we construct comparable 999 replicate weights (see app. A.2).

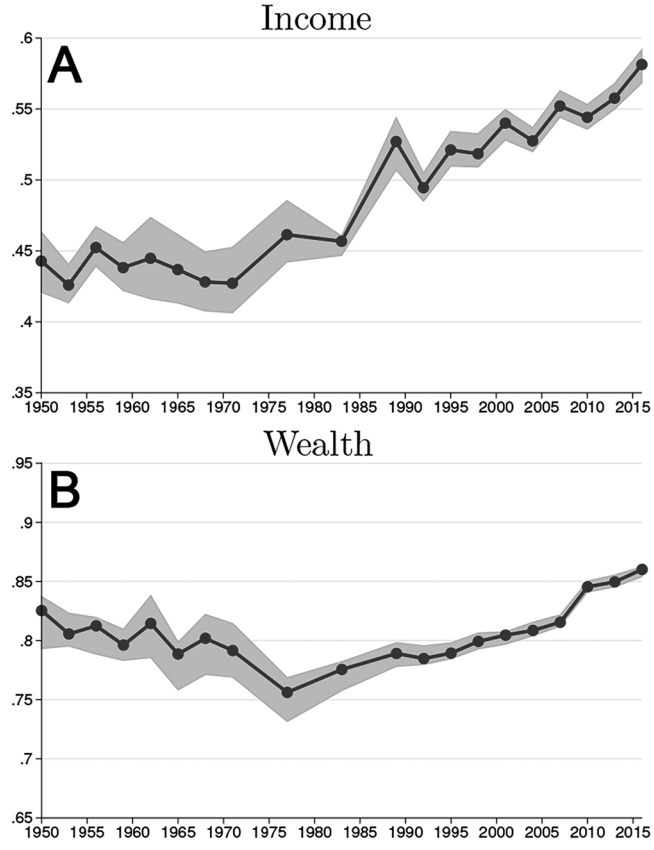


FIG. 4.—Gini coefficients for income (*A*) and wealth (*B*) with 90% confidence bands (shaded areas). Confidence bands are bootstrapped using 999 different replicate weights constructed from a geographically stratified sample of the final data set. A color version of this figure is available online.

B. Income and Wealth Shares

Decomposing inequality trends, we start with an exploration of changes in income and wealth shares at the top, following the recent literature.[14] Broadly speaking, the SCF+ data corroborate the trajectories of the US income and wealth distribution that emerged from the well-known studies by Piketty and Saez (2003) and Saez and Zucman (2016).

[14] We follow the recent literature in considering synthetic income and wealth groups. Households from a group need not be the same across surveys because of mobility. Exploring wealth mobility using data from the Panel Study of Income Dynamics, we find high persistence of households within wealth groups. More than 70% of households typically remain within wealth groups between survey dates. We report detailed results in app. C.

Figure 5*A* compares the income shares of the top 10% and 5% of the income distribution in the SCF+ with those calculated by Piketty and Saez (2003) using Internal Revenue Service (IRS) data.[15] Figure 5*B* compares top wealth shares from the SCF+ with those from Saez and Zucman (2016). Figure 5 also shows estimated 90% confidence bands resulting from sampling error in the SCF+ data for the top income and wealth shares. The confidence bands underscore that the reported increases in income and wealth inequality are statistically significant. Despite some minor discrepancies, the SCF+ and tax data align in levels and trends of inequality so that they tell a similar story about the long-run trajectory of wealth and income inequality in postwar America.

The 1962 SFCC constitutes an alternative data point to our SCF+ data for that year. The survey shows a nearly identical top 10% wealth share and a somewhat lower top 10% income share. This implies that the increase in income concentration at the top since the 1960s is even stronger when using the 1962 SFCC data point. However, the income tax data are actually much closer to the SCF+ data and also show a higher top 10% share than the SFCC. For consistency reasons, we use the SCF+ data throughout.

In the next step of our decomposition, we move down the distribution and turn to the evolution of income and wealth among the bottom 90% (table 4). The mirror image of increasing concentration of income in the hands of the top 10% must, by definition, be (relative) income losses among the bottom 90%. But which strata of the bottom 90% were hit particularly hard by the growing income share of the top 10%?

Table 4 reports the income shares of different groups of the income distribution and wealth shares of different strata of the wealth distribution.[16] Starting with income (cols. 1–5), the SCF+ shows that the top 10% have grown their income share by close to 15 percentage points from 34.5% to 47.2% between 1950 and 2016. The income share of the bottom 50% of Americans has fallen by roughly one-third from 21.6% to 14.6%, and middle-class households (50th–90th percentiles) have lost about 6 percentage points in income shares. In other words, we do observe a hollowing out of middle-class America, with households around the median having witnessed the largest relative income losses.

Columns 6–10 of table 4 show the change in wealth shares (households are now stratified by wealth). The main insight here is that until the 2008 financial crisis, changes in wealth shares were modest. If anything, the bottom 90% wealth share was slightly higher in 2007 than it was in 1950 and very close to its 1971 level. In contrast to the observed

[15] Piketty and Saez (2003) include salaries and wages, small business and farm income, partnership and fiduciary income, dividends, interest, rents, royalties, and other small items reported as other income.

[16] Appendix E.3 reports the full time series.

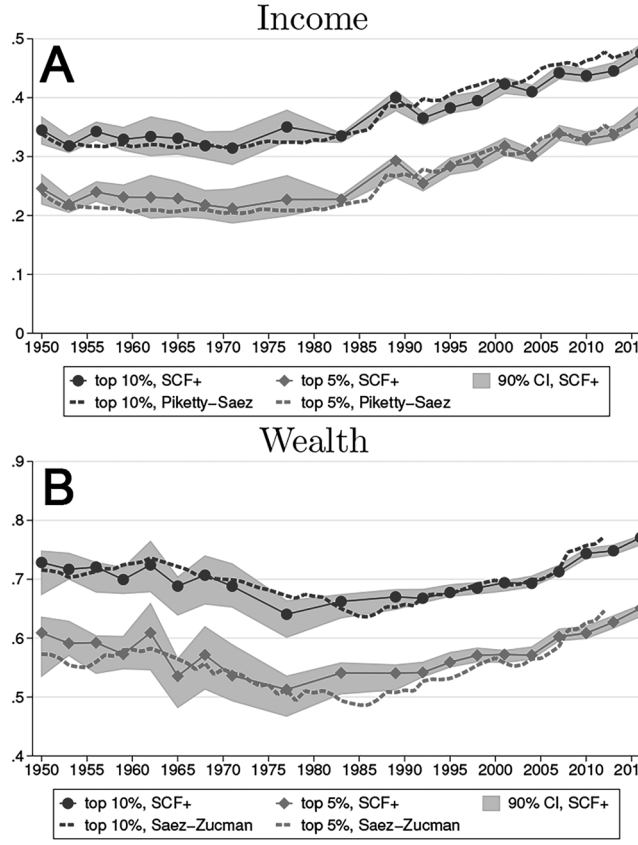


FIG. 5.—Top 5% and top 10% income and wealth shares from SCF+ data, Piketty and Saez (2003), and Saez and Zucman (2016). *A*, Income shares. Circles show the top 10% income shares from SCF+ data, and dark gray dashed line shows the top 10% income shares from Piketty and Saez (2003) using IRS tax data. Diamonds show the top 5% income shares from SCF+ data, and light gray dashed line shows the top 5% income shares from Piketty and Saez (2003). Shaded areas show 90% confidence bands for SCF+ estimates. *B*, Wealth shares. Circles show the top 10% wealth shares from SCF+ data, and dark gray dashed line shows the top 10% wealth shares from Saez and Zucman (2016) using IRS data and the capitalization method. Diamonds show the top 5% wealth shares from SCF+ data, and light gray dashed line shows the top 5% wealth shares from Saez and Zucman (2016). Shaded areas show 90% confidence bands for SCF+ estimates. A color version of this figure is available online.

changes in the income distribution, middle-class households managed to maintain their wealth shares until the mid-2000s. The 50%–90% wealth share was higher in 2007 than in 1950 and only slightly lower than in 1989. It is equally clear that the financial crisis had a substantial effect on the wealth distribution. Middle-class wealth shares collapsed across the board, while the wealth share of the top 10% surged by 6 percentage

TABLE 4
SHARES IN AGGREGATE INCOME AND WEALTH

	INCOME					WEALTH				
DISTRIBUTION	1950 (1)	1971 (2)	1989 (3)	2007 (4)	2016 (5)	1950 (6)	1971 (7)	1989 (8)	2007 (9)	2016 (10)
Bottom 50%	21.6	21.6	16.3	15.5	14.6	2.3	3.3	3.0	2.5	1.1
0%–25%	6.0	6.2	5.0	4.5	4.5	−.4	.0	−.1	−.1	−.5
25%–50%	15.6	15.4	11.3	11.0	10.1	2.7	3.4	3.1	2.6	1.6
50%–90%	43.9	47.0	43.7	40.3	37.9	24.8	27.9	30.0	26.2	21.8
50%–75%	23.5	25.0	22.4	20.3	18.4	9.8	10.7	11.9	10.3	7.4
75%–90%	20.4	22.0	21.3	20.0	19.5	15.0	17.2	18.1	15.9	14.4
Top 10%	34.5	31.4	40.0	44.2	47.5	72.8	68.8	67.0	71.3	77.1

NOTE.—Distribution shows household groups of the income and wealth distribution. Columns 1–5 show the share in total income for each household group. Households are sorted by income. Shares of the bottom 50%, 50%–90%, and top 10% add to 100%. Shares for the bottom 50% and 50%–90% are further split into subgroups. Columns 6–10 show the corresponding wealth shares across household groups. Households are sorted by wealth.

points within less than a decade. The decade since the financial crises witnessed the largest spike in wealth concentration in postwar America.

The overall outcome was a more pronounced shift in the income distribution than in the wealth distribution since the 1970s. We return to this important fact in section IV. In the next step, we zoom in on the bottom 90% and study long-run distributional trends in the lower parts of the distribution as well as low- and negative-wealth households.

C. The Bottom 90%

Much of recent research on seminal trends in inequality has focused on developments at the very top of the distribution. This emphasis on the top 1% (and beyond) plays to the strength of the tax data that were, at least so far, the only source spanning the postwar decades on a continuous basis. However, the tax data can provide only a relatively coarse picture of developments in the lower parts of the distribution. The SCF+ fills this gap.

We start the analysis with income and wealth trends for percentiles across the bottom 90%. Figure 6*A* documents that income grew at a similar rate across the 25th, 50th, and 75th percentiles in the first two postwar decades. From the 1970s to the 1990s, the 25th and 50th percentiles experienced real income losses, while incomes at the 75th percentile stagnated. All groups saw a return to real income growth from the mid-1990s to mid-2000s, but only incomes at the 75th percentile have recovered from the 2008 crisis.

Looking at percentile ratios in figure 6*B*, we see that since 1980, income at the median evolved similarly to income at the 25th percentile so that the 50th–25th ratio did not change much over the past four

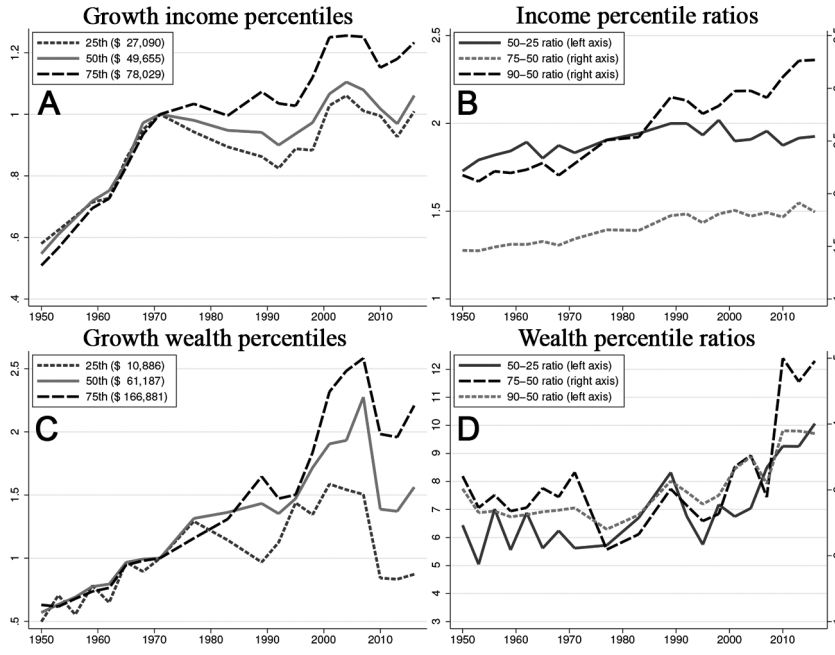


FIG. 6.—Percentile growth and percentile ratios for income and wealth. *A*, Growth of 25th, 50th, and 75th percentiles of income relative to 1971 (=1). Numbers in parentheses show levels of percentiles in 1971 (2016 dollars). *B*, 90th–50th, 75th–50th, and 50th–25th percentile ratios for income. *C*, Growth of 25th, 50th, and 75th percentiles of wealth relative to 1971 (=1). Numbers in parentheses show levels of percentiles in 1971 (2016 dollars). *D*, 90th–50th, 75th–50th, and 50th–25th percentile ratios for wealth. A color version of this figure is available online.

decades. By contrast, since the 1970s, the 75th and 90th percentiles left the median behind, leading to a pronounced widening of the 75th–50th and 90th–50th percentile ratios.

Figure 6*C* presents the same analysis for wealth. The picture is markedly different. First, wealth between groups started to persistently diverge only in the 2000s, not in the 1970s, as in the case of income. Second, households at all three percentiles saw major wealth drops after 2007, but there was considerable variation. The outcome is a substantial polarization of wealth and pronounced widening of the 90th–50th and 50th–25th ratios. Figure 6*C* also shows that nearly all wealth gains that households at the 25th percentile had made since 1971 have been wiped out by the crisis.

1. Low- and Negative-Wealth Households

Low- and negative-wealth households (net debtors) are key groups when it comes to the consequences of wealth inequality for macroeconomic

dynamics (Krusell and Smith 1998). Using the SCF+ data, we show in figure 7 how the shares of low- and negative-wealth households evolved over the past seven decades. The share of net debtors has doubled from its low during the 1980s but remains within its postwar range, which fluctuated between 5% and 12% (see fig. 7*A*). Starting in the 1980s, the average debt of net debtors increased from slightly less than 60% of average annual income in the period from 1950 to 1977 to over 140% in 2010. In 2016, the average debt balance of net debtors corresponds to 113% of their average income.[17] A broader measure of low-wealth households includes all households that have positive wealth but whose wealth is low relative to their income. We use a threshold of 3 months of income, implying a wealth-to-income ratio of 0.25 or below. This group can self-insure only to a limited extent by accessing savings, for instance, in the case of a job loss. The share of this group is large: close to one-quarter of American households are low-wealth households according to this definition. The share of these households has risen since the crisis but remains within its postwar range. One reason why households have negative wealth is negative home equity, and figure 7*B* reports the share of homeowners among negative-wealth households. The ratio reached its all-time high in 2007 when house prices collapsed, and highly leveraged households ended up *underwater*.

2. Wealthy Hand-to-Mouth Households

Kaplan and Violante (2014) argue that the group of households that behave like hand-to-mouth consumers—that is, as if they had no wealth for consumption smoothing—is much larger, as many households hold wealth in illiquid assets that cannot be easily accessed. Kaplan and Violante (2014) coined the term *wealthy hand-to-mouth households* and documented that from 1989 to 2010, about one in three American households can be classified as hand-to-mouth and that about two-thirds of these households are wealthy hand-to-mouth consumers.

Using the SCF+, we can provide estimates for the share of wealthy hand-to-mouth consumers for the entire postwar period. We follow Kaplan, Violante, and Weidner (2014) in identifying hand-to-mouth households in the data and relegate details to appendix A.7. We also provide estimates for cash holdings of households going back until 1973 using data from the National Crime Victimization Survey (NCVS).[18]

Figure 8 provides two different estimates of the share of hand-to-mouth and wealthy hand-to-mouth households in the United States over

[17] Results on average debt of net debtors are available from the authors upon request.

[18] Although the survey is designed to collect data on victims of crime, it also records details of the incidence, including theft. Appendix A.7 provides details on the construction of estimates, and fig. A.2 shows cash estimates as a fraction of median SCF+ income.

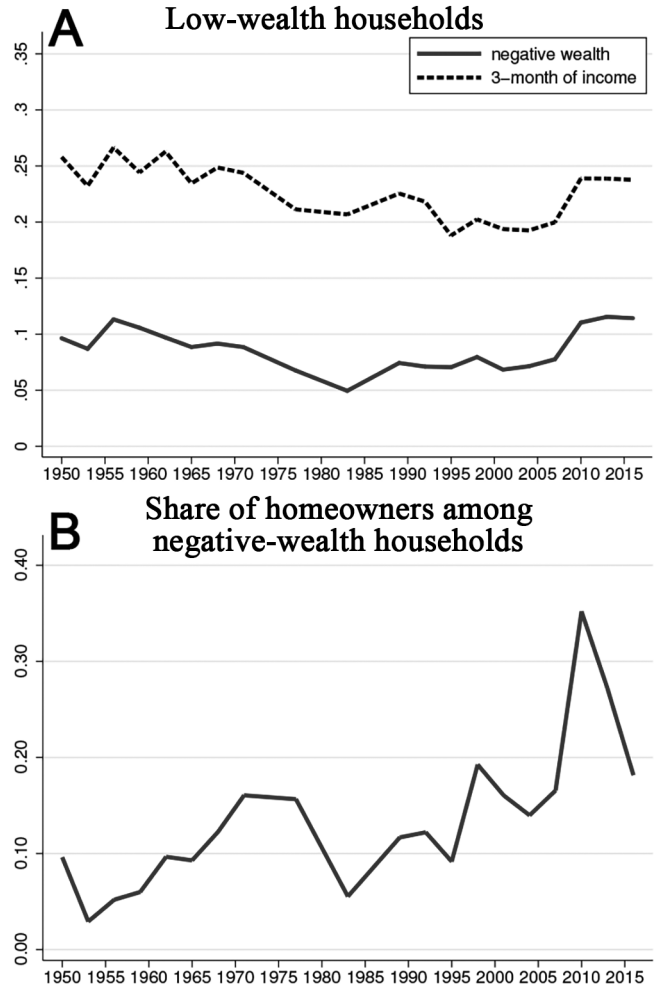


FIG. 7.—Low-wealth households. This figure shows the share of low-wealth households and the share of homeowners among negative-wealth households (net debtors). *A*, Shares for two measures of low-wealth households. Dashed line shows the share of households with wealth less than 3 months of income (3/12 of annual income). Solid line shows the share of negative-wealth households. *B*, Share of homeowners among households with negative wealth. A color version of this figure is available online.

the post–World War II period.[19] Figure 8*A* shows the baseline estimate following the approach for cash holdings in Kaplan, Violante, and Weidner (2014). It shows a slight downward trend for hand-to-mouth households

[19] We do not provide estimates for 1977 because income in 1977 is reported in intervals so that the share of hand-to-mouth is estimated imprecisely.

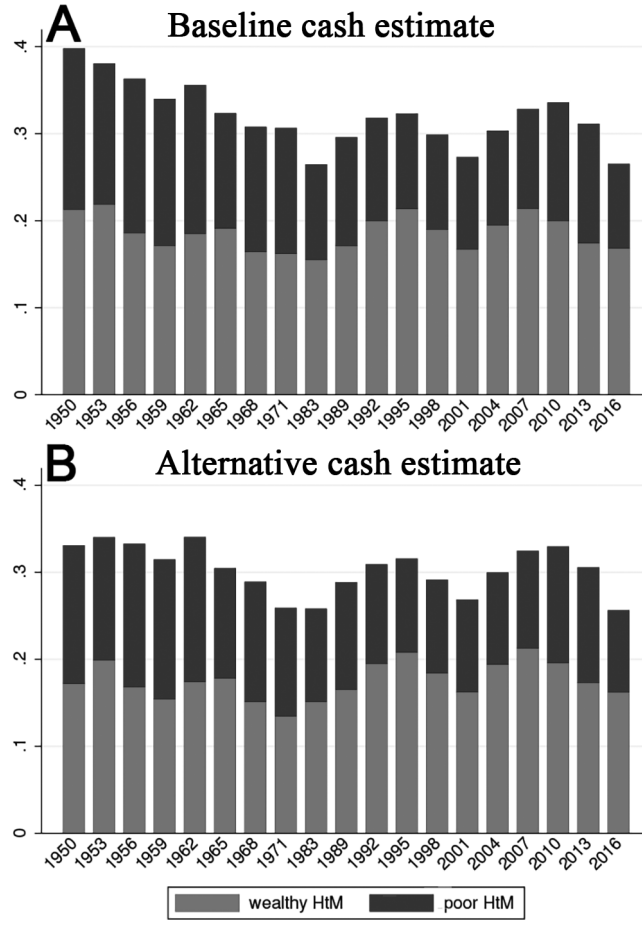


FIG. 8.—Shares of poor and wealthy hand-to-mouth households (HtM) for two alternative cash estimates. Bars show the total share of hand-to-mouth households in the population. Light gray bars show the share of wealthy hand-to-mouth households, and dark gray bars show the share of poor hand-to-mouth households. *A*, Estimated shares using cash estimates following Kaplan, Violante, and Weidner (2014). *B*, Estimated shares using cash estimates based on NCVS data. Estimates for 1977 are omitted because of data limitations (see n. 19). A color version of this figure is available online.

over the time period of the historical data and a rising proportion of wealthy hand-to-mouth households. Figure 8*B* provides estimates for hand-to-mouth households using our estimates for cash holdings from the NCVS data. Both panels show a relatively stable ratio of wealthy hand-to-mouth households since 1950, albeit with some variation over shorter horizons.

D. Demographic Change

What were the effects of secular changes in terms of educational attainment, age structure, and household size of the US population on income and wealth inequality? Using the demographic information in the SCF+, we provide answers to these questions. In a first step, we implement an approach proposed by Fortin, Lemieux, and Firpo (2011) to remove changes in the age structure and educational attainment over time.[20] In a second step, we account for changes in household size by adjusting income and wealth at the household level to per-adult equivalents using the Organization for Economic Cooperation and Development (OECD) equivalence scale.[21]

Figure 9*A* shows Gini coefficients for the original data and the two counterfactual cases where we add the marginal effects from fixing educational attainment and the age structure at the 1971 distributions. The effect from shifts toward more highly educated household heads on income appear rather small, but the effects coming from an older population are more sizable. Note that this finding is in line with a rising college wage premium, as we consider the effect only from changes in quantities (number of households), not prices (wages). In the case of wealth (fig. 9*B*), the effect of changing educational attainment and aging is small.[22] All in all, demographic changes have some effects but do not change the overall pattern of income and wealth inequality in the United States since World War II.

A second secular trend in the United States has been the decrease in average household size from an average of 3.4 household members in 1949 to an average of 2.5 in 2013, according to census data. Given that the SCF+ is a household survey, changes in household size can potentially affect measures of household-level inequality. We adjust income and wealth to per–adult equivalent member of the household with the OECD equivalence scale. Figure 9*C* reports that income concentration at the top falls somewhat when we look at adult equivalent income. This trend is consistent with stronger assortative mating and increasing female labor force participation. For wealth (fig. 9*D*), we do not observe big effects.

[20] We use 1971 as our base year for which we fix the distribution of demographic characteristics. We then estimate a probit model including age, educational attainment, the number of adults and children in a household, and the race of the household head as controls to derive adjustment weights. We relegate a detailed description to app. A.6.

[21] The OECD equivalence scale assigns a value of 1 to the first household member, 0.7 to each additional adult, and 0.5 to each child (http://www.oecd.org/eco/growth/OECD-Note-EquivalenceScales.pdf).

[22] Bartscher, Kuhn, and Schularick (2020) provide a detailed analysis on the trends in the financial situation of college and noncollege households in the United States based on the SCF+ data.

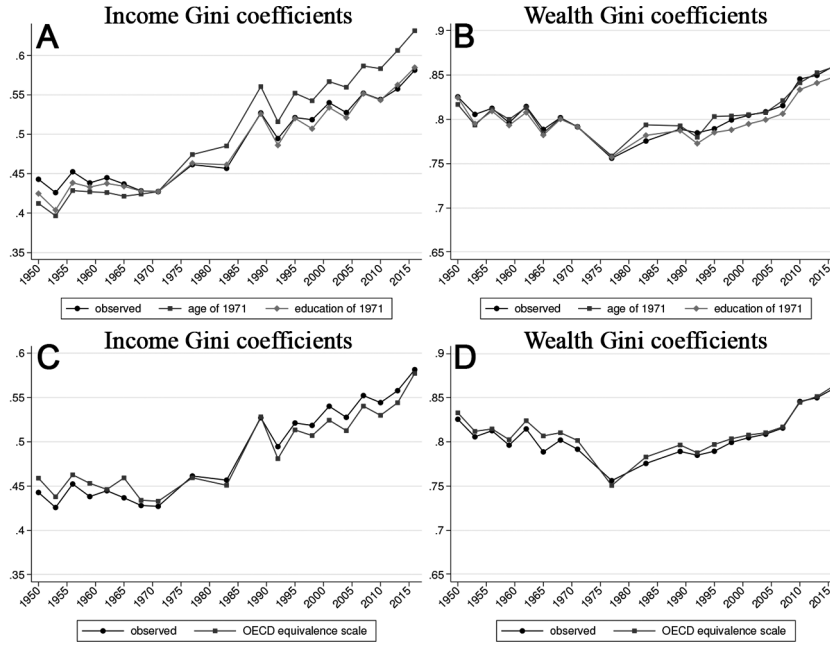


FIG. 9.—Gini coefficients for income and wealth accounting for changes in age, education, and household size composition over time. *A*, Three time series for Gini coefficients of income. Circles show the observed time series of Gini coefficients. Squares show counterfactual Gini coefficients for a constant 1971 age composition of households. Diamonds show counterfactual Gini coefficients for a constant 1971 education composition of households. Age and education refer to the head of household. *B*, Corresponding three time series for Gini coefficients of wealth. *C*, Two time series for Gini coefficient of income. Circles show the observed time series of Gini coefficients. Squares show Gini coefficients after dividing income by household size using the OECD equivalence scale (see n. 21). *D*, Corresponding two time series for Gini coefficients of wealth. A color version of this figure is available online.

E. The Persistence of Racial Disparities in Income and Wealth

Race is an important stratifying dimension of the US population. In a recent paper, Bayer and Charles (2017) provide long-run evidence on the black-white earnings gap using data from the US Census Bureau and the American Community Survey. They document persistent earnings differences for working-age men. The SCF+ data complement recent work on the long-run evolution of racial inequality along three dimensions.[23]

First, in addition to earnings, we study household income from all sources. Second, our unit of observation is the household, not working-age

[23] Thompson and Suarez (2017) and Dettling et al. (2017) analyze racial inequality using SCF data.

male individuals. We thus capture the effects of changing marriage patterns, higher labor force participation of women, as well as changes in transfers, education, and retirement decisions of households. Third, the SCF+ data also allow us to analyze the long-run evolution of wealth differentials between black and white households. So far, the racial wealth gap has remained uncharted territory, as long-run data were simply not available. With data reaching back to the pre–civil rights era, our analysis extends recent work by Wolff (2017), who studied wealth differences between black, white, and Hispanic households in the modern SCF data starting in 1983. For the analysis, we group households into black and white households according to the race of the household head.[24]

Figure 10 shows the trends in income and wealth of the median household and of the household at the 90th percentile for both white and black households. The racial divide will fall if black households' income or wealth increases more strongly over time. A lockstep evolution of the series for black and white households (equal growth rates) signals persistence of existing racial disparities.

Three facts stand out. First, income has grown at a comparable rate for black and white households. This means that pre–civil rights era disparities have largely persisted, as black income growth did not accelerate relative to white households. Second, as the numbers indicate, the size of the racial income divide remains substantial. The median black household has about half of the income of the median white household. Third, the wealth gap is much larger than the income gap and equally persistent. The median black household disposes of 12% of the wealth of a median white household. In the 1980s, the wealth of the median black household stood at about $13,000 in 2016 prices—equivalent to the value of a car. The median white household had about $115,000—corresponding to the value of a small house. Looking at the time trends in more detail, we note two periods when the racial disparities narrowed temporarily. In the 1970s, the income of the median black household grew about 20% faster than the income of the median white household. However, the trend reversed in the 1980s, when the share of black households headed by women increased strongly.[25] The 2000s are the second period in which the racial income gap narrowed somewhat for the median household.

[24] The number of interracial marriages is growing but remains small. Fryer (2007) reports that about 1% of marriages were interracial for whites and about 5% for black Americans. We drop all other racial categories. The survey questions for race in the SCF changed little over time. An important change happened in 1989, when the information was obtained as part of the interview rather than coded directly by the interviewer.

[25] When adjusting incomes for household size, the decline in relative incomes for black households during the 1980s becomes less pronounced.

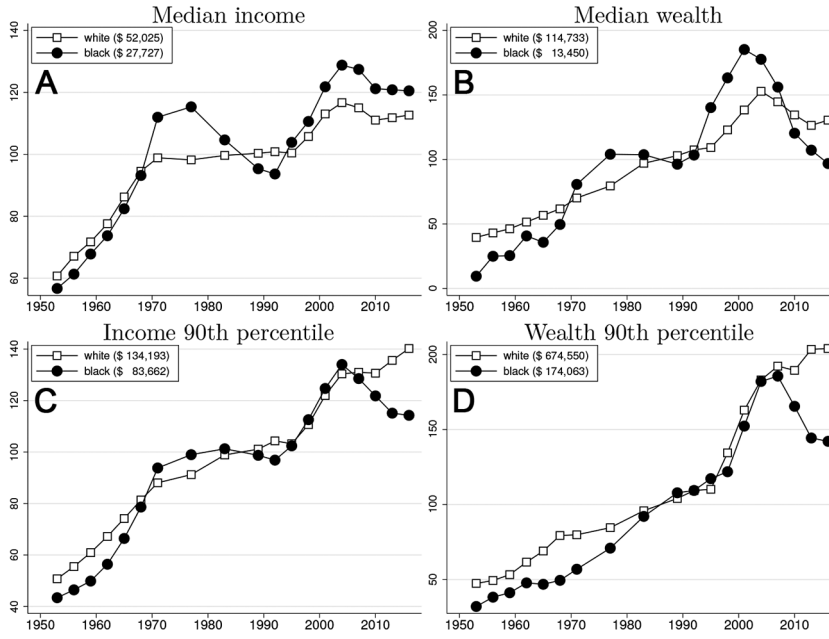


FIG. 10.—Trends of medians and 90th percentile of income and wealth for black and white households. *A*, Trends of median income for black and white households indexed to the period 1983–9 (=100). Numbers in parentheses are average income levels at the median over the indexing period (2016 dollars). *B*, Corresponding time series for median wealth. *C*, Trends at the 90th percentile of income for black and white households. *D*, Corresponding time series for 90th percentile of wealth. Time series are indexed to the period 1983–9 (=100). Numbers in parentheses are average income levels at the 90th percentile over the indexing period (2016 dollars). All time series show moving averages over three neighboring observations. Medians and 90th percentile always refer to the respective income and wealth distribution of black and white households.

Figure 10*B* exposes an equally persistent racial wealth gap. The difference in wealth narrowed temporarily in the housing boom of the 1990s and early 2000s but widened again after the financial crisis. After 2007, the wealth levels of households at the 90th percentile of the black wealth distribution collapsed, while the 90th percentile of the white wealth distribution remained largely unaffected.

As an alternative to study the evolution of earnings differences over time, Bayer and Charles (2017) apply the concept of a racial rank gap. Adapted for wealth, the rank gap is the percentage point difference between the rank of a given percentile in the black and white wealth distribution. For instance, a number of −30 for the median of the black wealth distribution means that the place of that household would be 30 percentage points lower in the wealth distribution of white households, that is, only at the 20th percentile.

Figure 11*A* shows the wealth rank gap at the median and the 90th percentile. For the median, the long-run average is close to −30, implying that the median black household is at the 20th percentile of the wealth distribution of white households. Put differently, the typical black household is poorer than 80% of white households. The rank gap fluctuates, tracking what we have seen for levels in figure 10*B*, but is highly persistent over time. We find an equally large and persistent rank gap at the

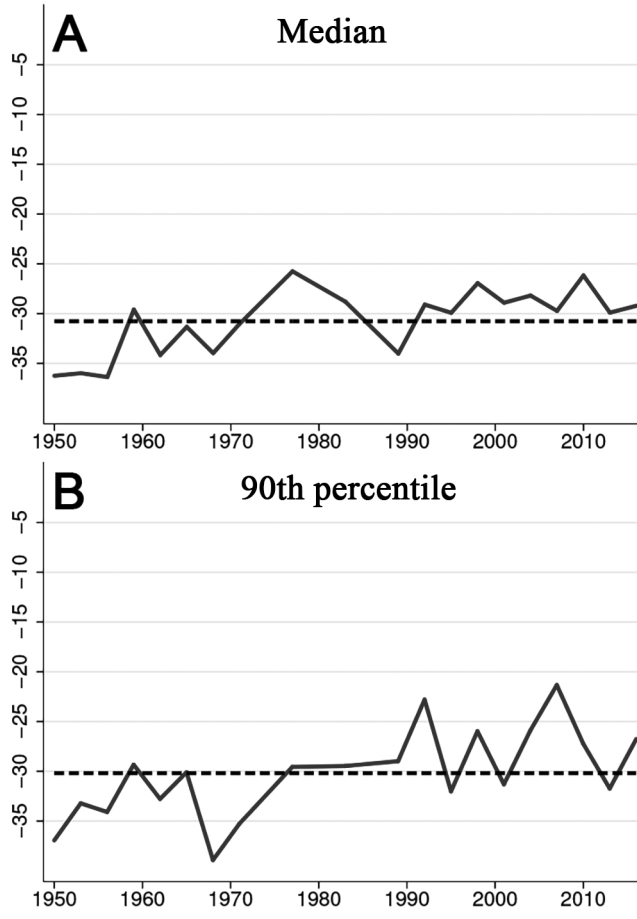


FIG. 11.—Racial rank gaps for wealth at the median and 90th percentile. *A*, Racial rank gap at the median. The racial rank gap is the difference in percentage points between the rank that the wealth level of the median black household takes in the wealth distribution of white households and the rank of the median white household. Dashed line shows the long-run average of the racial wealth rank gap. *B*, Corresponding racial rank gap at the 90th percentile. Dashed line shows the long-run average of the racial wealth rank gap. A color version of this figure is available online.

90th percentile of the wealth distribution. Our main conclusion is that virtually no progress has been made over the past 70 years in reducing wealth inequality between black and white households.

IV. Asset Prices and the Wealth Distribution

In section III, we discussed changes in the income and wealth distributions separately, as in the existing literature. Yet it is precisely the link between the income and wealth distributions that plays a central role in theoretical models of wealth inequality. A central advantage of the SCF+ is that it allows us to study the long-run evolution of the joint distribution of income and wealth. This topic is what we turn to now.

In the simplest model of the dynamics of the wealth distribution, changes in the income and wealth distributions are closely linked. With saving rates that are constant over time and uniform across wealth classes, and with uniform returns on wealth along the wealth distribution, changes in the wealth distribution would be solely driven by changes in the income distribution. Or, put differently, the differential growth rates of wealth would be a function of the differential growth rates of income. Recent studies have questioned this assumption, as models based on labor income risk typically produce too little wealth concentration at the top and cannot account for substantial shifts in wealth inequality that occur over short time horizons (Gabaix et al. 2016; Hubmer, Krusell, and Smith 2017; Benhabib and Bisin 2018).

As a first check, in figure 12 we compare the time path of income and wealth growth in the United States since 1971. Note that we stratify all households by wealth and index income and wealth levels to 1 in 1971. Figure 12*A* highlights a substantial divergence in income growth for different groups of the wealth distribution. Income growth was low for the bottom 90% and particularly meager for households in the lower half. For the bottom 50%, real incomes have stagnated since the 1970s. For households between the 50th and 90th percentiles of the wealth distribution, real incomes rose modestly by about one-third over nearly 40 years, implying annual growth rates of much less than 1% per year. By contrast, income growth at the top was strong. The incomes of households within the top 10% of the wealth distribution have doubled between 1971 and 2007.[26]

Yet when we turn to wealth growth for the same groups in figure 12*B*, the contrast is stark. From 1971 to 2007 (the last precrisis survey), wealth growth has been, by and large, identical for the top 10% and the bottom 90% of the wealth distribution. More precisely, middle-class (50%–90%)

[26] Appendix E.3 reports income shares for households along the wealth distribution comparable to the income shares along the income distribution in table 4.

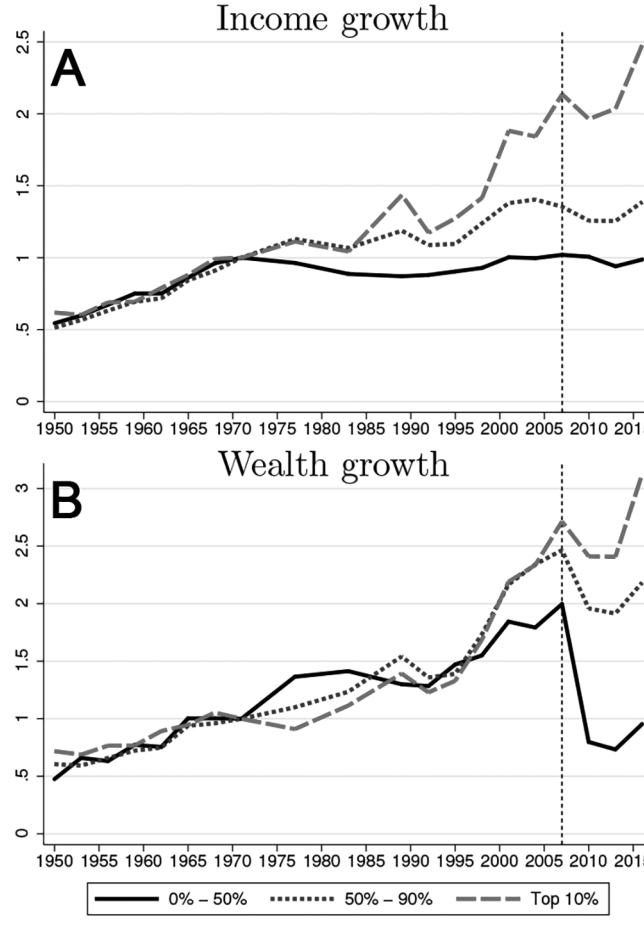


FIG. 12.—Income and wealth growth for different groups along the wealth distribution. *A*, Income growth for three groups in the wealth distribution: the bottom 50% (solid line), the middle class 50%–90% (dotted line), and the top 10% (dashed line). All income time series are indexed to 1 in 1971. *B*, Wealth growth for the same three groups along the wealth distribution. Vertical lines in both panels indicate the 2007 survey. A color version of this figure is available online.

wealth increased by 140% the same rate as top 10% wealth. And even the bottom 50% did not do too badly when it comes to wealth growth, as their wealth still doubled between 1971 and 2007. Wealth and income growth rates have decoupled over an extended period, in marked contrast to the simple model sketched above. We will return to this point below.

Figure 12*B* also shows how devastating the 2007–8 financial crash was to lower middle-class wealth, while the impact of the crisis on wealth at

the top was rather minor. By 2013, the absolute level of real wealth below the median household was 20% below its 1971 level. Within a few years, the crisis wiped out all gains in household wealth that the bottom 50% of the distribution had made over the preceding four decades. As of 2016, still close to half of the American population dispose of less wealth in real dollar amounts than in 1971.[27]

One upshot is that before the financial crisis, wealth-to-income ratios increased most strongly in the middle and at the bottom of the wealth distribution and then also fell most strongly for those groups during the crash. Figure 13 illustrates this phenomenon. Figure 13*A* shows the strong increase in wealth-to-income ratios for the bottom 50% (left *Y*-axis) and the middle class (right *Y*-axis) until 2007, followed by a substantial decline. Figure 13*B* shows wealth-to-income ratios for the top 10% that are much higher on average. Wealth-to-income ratios increased only slightly between 1971 and 2007 at the top and hardly changed after 2007.

A. The Dynamics of the Wealth Distribution

If we are to understand the dynamics of the wealth distribution in America over the past seven decades, we must look beyond income growth. In the following, we demonstrate that asset price changes played an important role in the observed dynamics of wealth inequality in postwar America.[28]

Asset prices affect the dynamics of the wealth distribution through two channels. First, asset prices lead to differential capital gains if portfolios differ across the distribution. We document this important stylized fact for the US economy below. As changes in asset prices revalue existing wealth, they can induce shifts in wealth shares that are unrelated to income changes. Moreover, they can do so over short horizons, as they immediately affect the value of accumulated assets. We will document that in America, persistent differences in portfolio composition between middle-class households and rich households essentially give rise to a race between the stock market and the housing market in shaping the dynamics of the wealth distribution.

The second channel through which asset prices matter for the dynamics of wealth inequality is through their effect on wealth-to-income ratios. The level of the wealth-to-income ratio determines the relative importance of savings flows for wealth dynamics. When wealth-to-income ratios

[27] We provide a sensitivity analysis to these growth trends when using the PCE instead of the CPI for inflation adjustment in app. D.4. See n. 7 for further discussion.

[28] For the French case, Garbinti, Goupille-Lebret, and Piketty (2017) show that price effects played an important role in shaping the French wealth distribution over the past 200 years. In the American context, Saez and Zucman (2016) discuss that price effects can change inequality trends relative to those implied by income and saving rate differences but focus on saving rate differences in their discussion.



FIG. 13.—Wealth-to-income ratios by wealth group, 1950–2016. *A*, Wealth-to-income ratios for the bottom 50% of the wealth distribution (solid line) and for the 50%–90% of the wealth distribution (dotted line). *B*, Wealth-to-income ratio for the top 10% of the wealth distribution. Wealth-to-income ratios are constructed as the ratio of averages within wealth groups. A color version of this figure is available online.

are high, income growth and savings flows become relatively less important for the wealth distribution, simply because the stock of wealth is high relative to income flows. This second channel is particularly relevant for the time period from 1970 to 2007, when the aggregate wealth-to-income ratio increased from 4 to more than 7, driven by rising house prices and a booming stock market.

To see how asset prices affect wealth inequality when portfolios are heterogeneous, consider a household i in period t with a portfolio of assets $\{A^i_{j,t}\}^J_{j=1}$, for instance, houses, stocks, and saving accounts. For household i, the capital gain Π^i_t from asset price changes between period t and $t+1$ is the asset-weighted average of price changes

$$\Pi^i_t = \sum_{j=1}^{J} \left(\frac{p_{j,t+1}}{p_{j,t}} - 1 \right) A^i_{j,t},$$

where $p_{j,t}$ denotes a (real) price index for asset j in period t. Denote the household's wealth in t by W^i_t and divide both sides of the equation by wealth to obtain

$$\begin{aligned} \frac{\Pi^i_t}{W^i_t} &= \sum_{j=1}^{J} \left(\frac{p_{j,t+1}}{p_{j,t}} - 1 \right) \frac{A^i_{j,t}}{W^i_t}, \\ q^i_t &= \sum_{j=1}^{J} \left(\frac{p_{j,t+1}}{p_{j,t}} - 1 \right) \alpha^i_{j,t}, \end{aligned} \tag{1}$$

where $\alpha^i_{j,t}$ denotes the portfolio share $A^i_{j,t}/W^i_t$ of asset j for household i in period t and q^i_t is the growth rate of household wealth from capital gains. Equation (1) shows that portfolio differences (i.e., differences in $\alpha^i_{j,t}$ across households) lead to differences in capital gains q^i_t.

To fix ideas and structure the discussion about how this affects the wealth distribution, we rely on an illuminating accounting framework adapted from Saez and Zucman (2016).[29] Consider a simplified law of motion for wealth of household i:

$$W^i_{t+1} = W^i_t(1 + r^i_t + q^i_t) + Y^i_t - C^i_t,$$

where r^i_t are returns on wealth other than capital gains (e.g., dividends), Y^i_t denotes income from all other sources, and C^i_t denotes consumption.[30] The savings flow S^i_t of household i in period t corresponds to total income net of consumption $S^i_t = r^i_t W^i_t + Y^i_t - C^i_t$. Define further the saving rate s^i_t as $s^i_t = S^i_t/Y^i_t$, so that the law of motion for wealth becomes

[29] A micro-founded analysis of household saving behavior and portfolio choice requires a more complex approach. Hubmer, Krusell, and Smith (2017) discuss why such a framework remains beyond reach for the time being but must become a topic for future research.

[30] Income denotes income from all sources excluding capital. This approach simplifies wealth dynamics by abstracting from bequests and death, divorce, marriage, or other life-cycle events that affect households' wealth accumulation. We adjust the framework by Saez and Zucman (2016) slightly with respect to the timing convention by assuming that capital gains accrue together with savings flows, but the underlying mechanism remains the same. Saez and Zucman (2016) focus on the heterogeneity in savings behavior and assume homogeneity of capital gains. Our discussion focuses on the comparison of the relative importance of savings and capital gains as drivers of wealth accumulation.

$$W^i_{t+1} = W^i_t(1 + q^i_t) + S^i_t = W^i_t(1 + q^i_t) + s^i_t Y^i_t = (1 + q^i_t + \sigma^i_t)W^i_t, \quad (2)$$

with $\sigma^i_t = s^i_t Y^i_t / W^i_t$ capturing the contribution of savings to wealth growth and the term q^i_t capturing the effect of capital gains to wealth growth.

In the next step, we move from the law of motion for wealth levels to a law of motion for wealth shares of different wealth strata. We construct synthetic saving flows and capital gains for specific wealth groups, again taking the lead from Saez and Zucman (2016). Savings flows and capital gains for wealth groups are synthetic in the sense that they assume that households stay in their wealth group from one period to the next. Using data from the Panel Study of Income Dynamics, we show in appendix C that while there is some mobility in practice, the synthetic method yields a good approximation of wealth dynamics.

Note that we will now use i to refer to the group of households in a specific wealth stratum. The wealth share of group i in period t is $\omega^i_t = W^i_t / W_t$, where W_t is aggregate wealth in period t. All aggregate variables are defined according to group-level variables; for example, the aggregate savings rate is $s_t = S_t / Y_t$. Applying some straightforward transformations to equation (2) yields the law of motion for the wealth share ω^i_t:

$$\omega^i_{t+1} = \frac{1 + q^i_t + \sigma^i_t}{1 + q_t + \sigma_t}\omega^i_t \Leftrightarrow \frac{\omega^i_{t+1}}{\omega^i_t} = \frac{1 + q^i_t + \sigma^i_t}{1 + q_t + \sigma_t}. \quad (3)$$

The law of motion has an intuitive interpretation: the wealth share of any group i increases if the group's wealth growth rate exceeds the average wealth growth rate in the economy. Differences in growth rates result from the two components of wealth growth in equation (2). First, group i's capital gains q^i_t can be higher (or lower) than the average capital gain q_t in the economy. Second, different rates of wealth growth can result from the difference between group i's savings component σ^i_t relative to the average savings σ_t. This is the channel through which differences in income growth translate into wealth inequality: higher incomes of group i will, all else equal, increase the group's saving flows and its wealth level relative to other groups in the economy.

The savings term σ^i_t comprises the inverse of the wealth-to-income ratio. With higher wealth-to-income ratios, the importance of savings flows declines and tends to zero. This implies that the relative importance of differences in savings flows diminishes with higher wealth-to-income ratios. This effect is independent of portfolio composition and is operative even when households have identical wealth portfolios (so that capital gains are identical). Also note that the two components q^i_t and σ^i_t are independent, so that a low savings component relative to the average can go hand in hand with a large capital gains component relative to the

average for group *i*, and vice versa. This can decouple the evolution of the income and wealth distributions.

B. Portfolio Heterogeneity and Asset Price Exposures

If portfolios differ systematically along the wealth distribution, asset price changes will lead to differential capital gains along the wealth distribution. These in turn can induce changes in the wealth distribution that are unrelated to changes in the income distribution. The necessary condition for such effects is that portfolios are heterogeneous. For the first time, the SCF+ provides us with long-run balance sheet information to study the composition of household portfolios over a long time horizon. The evidence points to systematic and highly persistent differences in wealth portfolios across groups and hence a potentially important role for asset prices in shifting the wealth distribution, as we will now demonstrate.

Figure 14 displays the heterogeneity of household portfolios. It tracks the portfolio composition of the bottom 50%, the 50%–90%, and the top 10% of the wealth distribution since 1949. As a benchmark, we also track the average portfolio of the macroeconomy. In the figures, assets enter with positive values and debt as negative values. Household wealth corresponds to the consolidated value of all portfolio positions and is indicated by a dashed line in each of the figures. The degree of leverage in household portfolios can be inferred by looking at the sum of assets in excess of wealth. We provide time series in the appendix E.5.

Bottom 50%.—The bottom 50% have little wealth, yet their small wealth position masks substantial gross positions. Houses and other nonfinancial assets, mainly cars, make up more than 80% of the asset side of the balance sheet. Financial assets play a minor role in bottom 50% portfolios. On the liability side, housing debt is the dominant form of debt, but compared with other wealth groups, the bottom 50% also have a high share of nonhousing debt. In recent years, education loans make up a growing share of this debt (Bartscher et al. 2019). Assets exceed wealth by a large margin, indicating a high degree of leverage.

Middle class (50%–90%).—The middle-class portfolio is dominated by nonfinancial assets. About two-thirds of the middle-class portfolio consists of houses and other nonfinancial assets. Direct stock holdings are typically below 5%. The large growth of other financial assets in the portfolio comes mainly from defined-contribution pension plans. The middle class is also leveraged, with housing debt being the dominant debt component and assets exceeding wealth by 10%–30%.

Top 10%.—The top 10% are different when it comes to portfolio composition. The bulk of wealth is held in stocks and business equity. Houses as an asset class gained in importance for the top 10% but constitute a comparatively small fraction of assets. Other financial assets have grown

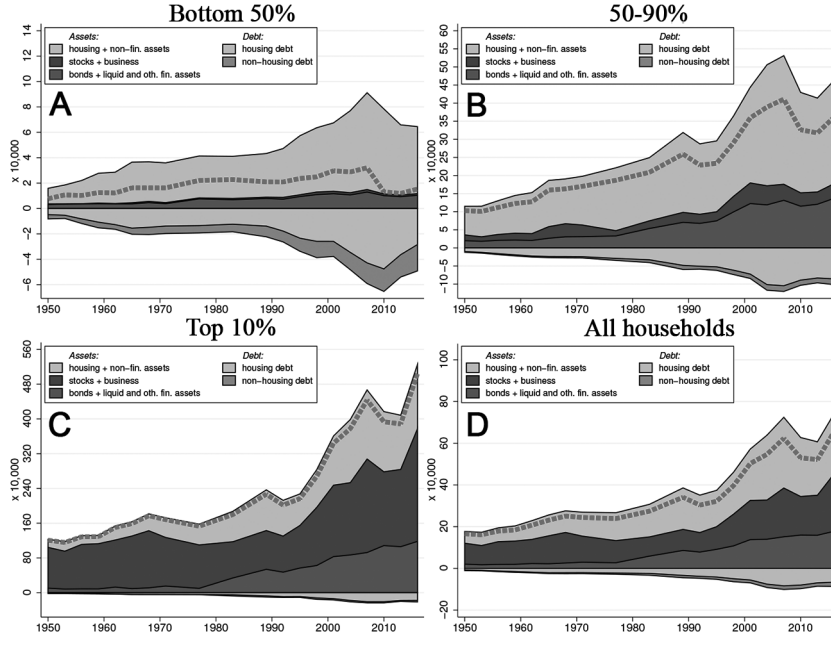


FIG. 14.—Heterogeneity in household portfolios for different groups along the wealth distribution. *A*, Portfolio composition of the bottom 50% of the wealth distribution. Assets are shown as positive portfolio components, and debt is shown as negative portfolio components. Dashed line indicates mean wealth of this group. All portfolio components and wealth levels are shown in $10,000 (2016 dollars). *B*, Equivalent portfolio of the middle class (50%–90%) together with mean wealth of this group (dashed line). *C*, Equivalent portfolio of the top 10% of the wealth distribution and average wealth for this group (dashed line). *D*, Equivalent portfolio for all households. Dashed line indicates mean wealth. Wealth groups are separately defined for each survey year. A color version of this figure is available online.

strongly, mainly because of the proliferation of defined-contribution pension plans. Leverage is low, so that for the top 10%, the sum of assets corresponds approximately to wealth.

Portfolio composition thus varies substantially along the wealth distribution. These differences are also highly persistent. The portfolios of the bottom 90% are nondiversified and highly leveraged. Houses are the asset of the bottom 90%, making residential real estate the most egalitarian asset. Figure 15 highlights this point by showing the ownership structure of housing and stocks at different points in time.[31] The bottom 90% hold about half of all housing wealth but only a tiny fraction of stocks. Stocks

[31] We include mutual funds in the stock holdings. Results change little if we consider only direct stock holdings.

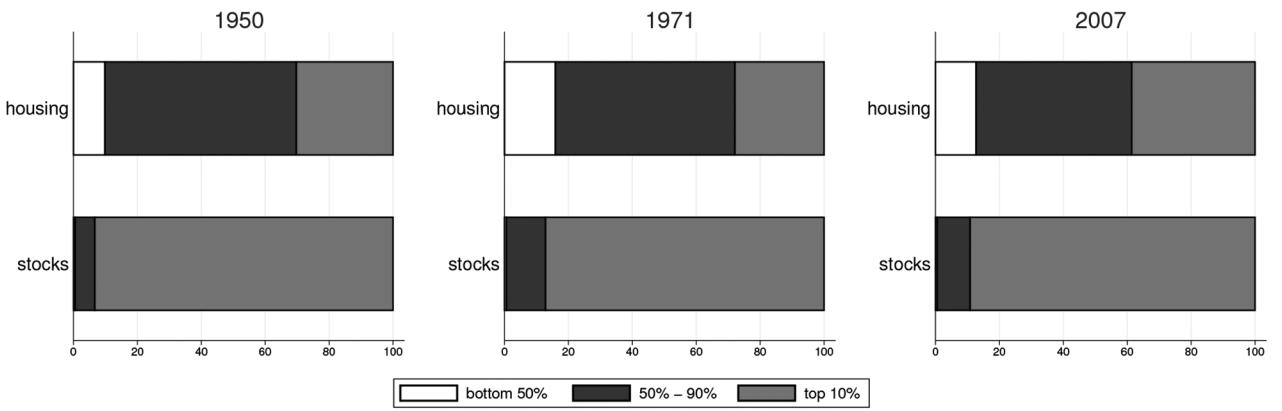


FIG. 15.—Shares in total asset holdings by wealth group for 1950 (*left*), 1971 (*middle*), and 2007 (*right*). Each bar shows the share of total assets held by different wealth groups. Wealth groups are the bottom 50%, the 50%–90%, and the top 10% of households. Top bars show housing assets; bottom bars show stocks, including mutual funds. In each bar, the white part shows the share of the asset in that year held by the bottom 50%, the dark gray part shows the share held by the 50%–90%, and the light gray part shows the share held by the top 10%. Shares across wealth groups for each asset and year add up to 100%. A color version of this figure is available online.

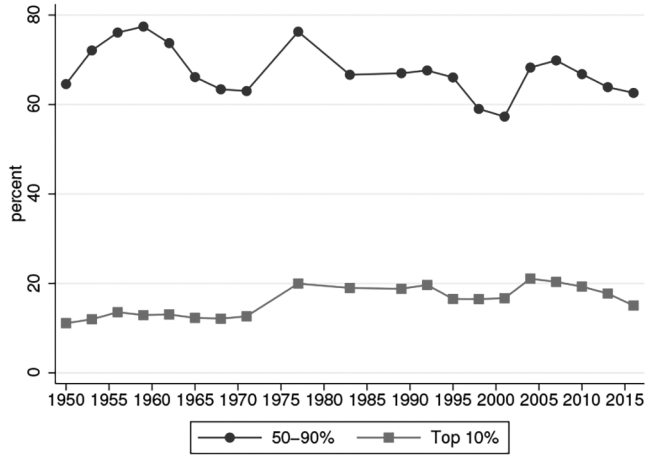


FIG. 16.—House price exposure (housing/wealth × 100) for different wealth groups. House price exposure is measured by the elasticity of household wealth with respect to house price changes (housing/wealth × 100). Circles show house price exposure for the 50%–90% of households in the wealth distribution, and squares show house price exposure for the top 10% of households in the wealth distribution. A color version of this figure is available online.

are the asset of the wealthy in the sense that the top 10% consistently hold more than 90% of stocks. Looking at the Gini coefficient for individual asset classes confirms that housing is the most equally distributed asset and that the distribution of housing wealth has not changed substantially over time. We report Gini coefficients for all asset classes in the appendix E.1.

An important consequence of nondiversified and leveraged portfolio positions is that the wealth of middle-class households is highly exposed to changes in house prices. We quantify this exposure as the elasticity of wealth with respect to house prices, which is equal to housing/wealth, the ratio of the asset value of housing to wealth. Figure 16 shows the resulting exposure to house prices for middle-class households and households in the top 10% over time. The figure confirms that the elasticity of middle-class wealth to house prices is three to four times higher than at the top. A 10% increase in house prices increases middle-class wealth by 6%–7%.[32] For changes in stock prices, the exposures are reversed. The top 10% are highly exposed, the rest very little.

[32] We do not show the bottom 50% in this graph because of their large exposure. Appendix E.2 provides further results on house price exposure along the wealth distribution and changes over time and decomposes the house price elasticity of wealth further into a *diversification component* and a *leverage component*.

C. *The Race between the Stock Market and the Housing Market*

Such pronounced portfolio differences between households along the wealth distribution give rise to what we call the race between the stock market and the housing market: owing to their larger exposure, the middle class gains relatively more than top-wealth households when house prices rise. All else equal, rising house prices make the wealth distribution more equal, while stock market booms have the opposite effect: they primarily boost wealth at the top and lead to a more unequal distribution of wealth.

To explore how important this race between the stock market and the housing market has been for the wealth distribution in postwar America, we estimate the following regression relating changes in the top 10% wealth share over the 3-year survey intervals to asset price movements:

$$\Delta \log(\omega_{t+1}^{\text{top10}}) = \beta_0 + \beta_h \Delta \log(p_{t+1}^{h}) + \beta_s \Delta \log(p_{t+1}^{s}) + \varepsilon_t,$$

where Δ is the first-difference operator, $\Delta x_{t+1} = x_{t+1} - x_t$, the superscript h denotes house prices, and the superscript s denotes stock prices. We use the Standard and Poor's 500 stock market index and the Case-Shiller house price index obtained from the latest version of the Macrohistory Database (Jordà, Schularick, and Taylor 2017; Knoll, Schularick, and Steger 2017).

Column 1 in table 5 reports the estimated coefficients for the baseline regression. The signs of the estimated coefficients demonstrate how the race between the housing market and the stock market shaped wealth dynamics: rising house prices are associated with a falling top 10% wealth share. Rising stock prices boost the top 10% wealth share. Note that in the baseline specification, the error term comprises all other effects related to differences in savings or wealth-to-income ratios. Wolff (2002) estimated

TABLE 5
RACE BETWEEN STOCK MARKET AND HOUSING MARKET

	(1)	(2)	(3)	(4)
β_h	−.062	−.071	−.084	−.073
β_s	.024	.026	.024	.026
θ^{top10}	No	Yes	No	Yes
Y/W	No	No	Yes	Yes
Observations	19	19	19	19
R^2	.06	.23	.21	.23

NOTE.—This table shows the regression of changes in the top 10% wealth share on asset price growth and controls. Growth rates are computed using log differences. θ^{top10} denotes the income share of the top 10% of households in the wealth distribution. Y/W denotes controls for the inverse of the wealth-to-income ratio of the top 10% of households in the wealth distribution and for the aggregate economy. All observations from the surveys from 1950 to 2016 are used for the regressions.

similar regressions relating the top 1% and top 5% wealth share to fluctuations in stock and house prices.

We control for these factors in additional regressions in columns 2–4 by adding the income share of the top 10% and changes in the ratio of income to wealth as regressors. The coefficients β_h and β_s become larger and are economically significant, but the sample is so small that they are not statistically significant.

The estimated regression coefficients have an intuitive interpretation, as they correspond to the average elasticity of the top 10% wealth share with respect to asset prices.[33] From the law of motion for the wealth share, we can derive the elasticity of the wealth share of group i with respect to the price of asset j:

$$\frac{\partial(\omega^i_{t+1}/\omega^i_t)}{\partial\left(p^j_{t+1}/p^j_t\right)} = (1 + q_t + \sigma_t)^{-1}\left(\alpha^i_{t,j} - \alpha_{t,j}\frac{\omega^i_{t+1}}{\omega^i_t}\right).$$

Figure 17 shows the time series for the elasticity constructed from the portfolio shares in the SCF+ data. The house price elasticity of the top 10% fluctuates around a mean of −0.17, reasonably close and within the confidence bounds of the point estimates in the wealth share regressions above. All else equal, an elasticity of −0.17 implies that a 10% increase in house prices will lower the top 10% wealth share by 1.7%. Assuming a top 10% wealth share of 75%, this corresponds to a decrease in the top 10% wealth share of 1.3 percentage points.

The magnitudes are large. A hypothetical 40% increase in real house prices would reduce the wealth share of the top 10% from 75% to 70%, bringing it back to its 1971 level. For stock prices, the long-run average elasticity stands at 0.036 and is slightly larger than the point estimates from the regression above. A 130% real increase in the stock market—comparable to the period between 1998 and 2007—increases the wealth share of the top 10% by about 6 percentage points. By the same token, the fall in the stock market in the 1970s contributed to the decline in wealth inequality in the 1970s that we discussed above. Because of the substantial portfolio heterogeneity along the wealth distribution, asset prices have first-order effects on the evolution of wealth inequality.

D. Wealth Gains from Asset Prices

The results of section IV.C demonstrate that over short horizons, asset price fluctuations are closely associated with changes in wealth shares. We now quantify the contribution of asset price changes to wealth accumulation of different groups over the past four decades. We concentrate

[33] The portfolio share $\alpha_h = H/W$ in fig. 16 corresponds to the elasticity of the wealth level.

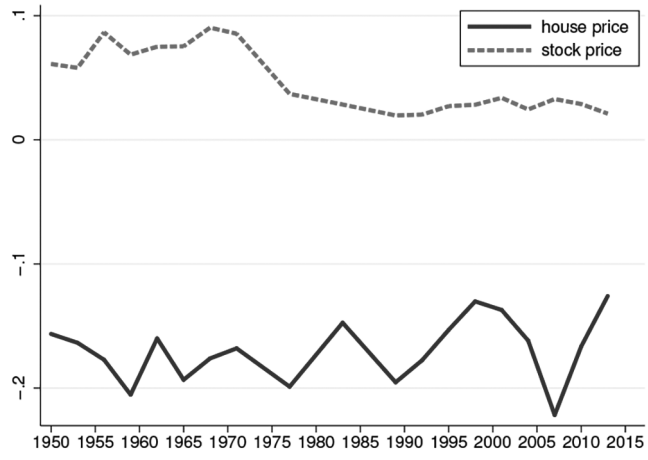


FIG. 17.—Elasticity of top 10% wealth share with respect to asset prices. Dashed line shows elasticity of the top 10% wealth share with respect to the stock price. Solid line shows elasticity of the top 10% wealth share with respect to the house price. A color version of this figure is available online.

on wealth growth over two distinct periods. The first period comprises the nearly four decades from 1971 to the 2007–8 financial crisis (the last precrisis survey was carried out in 2007). This was a period in which the income distribution widened substantially, but measures of wealth inequality changed very little. We will see that house price–induced wealth gains for the bottom 90% of the population played a central role in the observed stability of the wealth distribution.

The second period that we study covers the decade after the financial crisis. As discussed above, this decade has witnessed the largest increase in wealth inequality in postwar history. The income distribution, by contrast, changed only modestly over this period. We will see that asset prices are again central in accounting for the large shift in the distribution of wealth over a relatively short period.

Figure 18*A* shows how much wealth grew between 1971 and 2007 because of price changes in the stock market and the housing market. As before, we track wealth growth across three different wealth groups: the bottom 50%, the 50%–90%, and the top 10%. While figure 18*A* displays the wealth growth from asset price changes, figure 18*B* highlights the share of such asset price–induced wealth growth in total wealth growth, including changes in wealth coming from savings flows.

Two observations stand out. First, over the entire period, wealth gains from rising asset prices were substantial across the distribution. As figure 18*A* shows, the wealth of the bottom 50% grew by 90% only because of price effects. Also for the 50%–90% group and the top 10%, asset

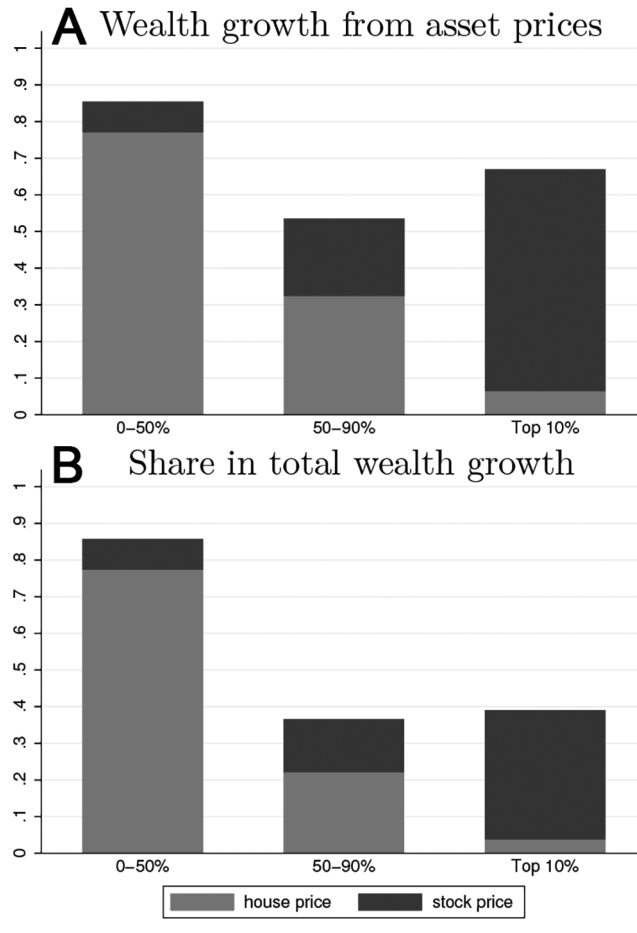


FIG. 18.—Wealth growth from changes in house prices and stock prices in levels and as share of total wealth growth for different wealth groups, 1971–2007. *A*, Wealth growth from asset price changes between 1971 and 2007 for the bottom 50%, 50%–90%, and top 10% of the wealth distribution. Wealth growth from asset prices is computed as asset price change multiplied by 1971 stock of asset holdings of wealth group divided by the 1971 wealth level. Wealth groups are the bottom 50%, 50%–90%, and top 10% of the wealth distribution. Wealth growth is shown relative to 1971 wealth levels. *B*, Share of wealth growth from asset prices from *A* as share in wealth growth of the respective group. Only asset price changes in housing and stock prices are considered. In each bar, the light gray part shows the contribution from house price changes, and the dark gray part shows the contribution from stock price changes. A color version of this figure is available online.

price changes induced sizable wealth growth of about 60%. It is easy to spot the race between the stock market and the housing market in the data: wealth growth from house price effects dominate for the bottom 90% of the wealth distribution, and stock price gains account for the bulk of the wealth gains at the top.

How much of the total wealth increase in the different groups is explained by price effects? Figure 18*B* shows that for the bottom 50%, virtually all wealth growth over the 1971–2007 period came from higher asset prices. But even in the middle and at the top, asset prices still account for 40% of total wealth growth, with the rest accounted for by savings. Note that these estimates for the role of asset prices for wealth growth are likely conservative, as households increased their exposure to the housing market over this period. The variation in the contribution to wealth growth along the distribution helps us to understand why wealth and income growth decoupled in the decades before the crisis: relative to income, wealth grew most strongly for the bottom 90%.

Asset price movements also explain why wealth concentration spiked after the financial crisis. House prices plummeted after 2007 and recovered only slowly in recent years. By 2016, they were still 10% below their 2007 peak level. By contrast, the stock market recovered more quickly, and the main stock market indices were about 30% above their 2007 levels in real terms in 2016. Figure 19 shows the race between the housing market and the stock market between 2007 and 2016. The bottom 50% lost 15% of wealth relative to 2007 levels, mainly because of lower house prices. By contrast, the top 10% were the main beneficiary from the

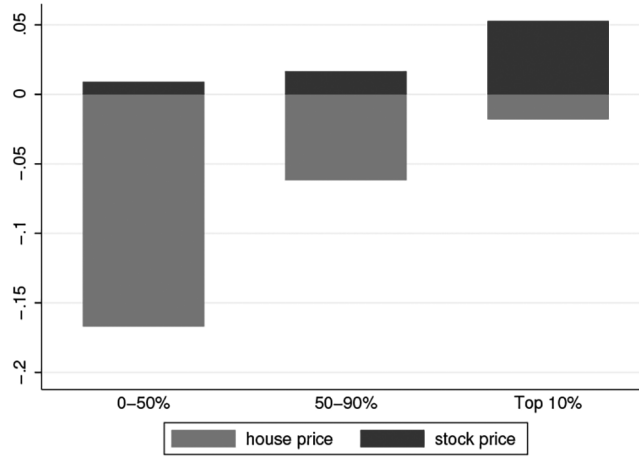


FIG. 19.—Wealth growth from changes in house prices and stock prices for different wealth groups between 2007 and 2016. Wealth groups are the bottom 50%, 50%–90%, and top 10% of the wealth distribution. Wealth growth from asset prices is computed as asset price change multiplied by 2007 stock of asset holdings of wealth group divided by the 2007 wealth level. Wealth groups are the bottom 50%, 50%–90%, and top 10% of the wealth distribution. Only asset price changes in housing and stock prices are considered. In each bar, the light gray part shows the contribution from house price changes, and the dark gray part shows the contribution from stock price changes. A color version of this figure is available online.

stock market boom and were relatively less affected by the drop in residential real estate prices. The consequence of substantial wealth losses at the bottom and in the middle of the distribution, coupled with wealth gains at the top, produced a large spike in wealth inequality.

What would the distribution of wealth in America look like today without asset price effects? To construct a counterfactual, we use the law of motion for wealth levels from equation (2), keeping all parameters constant but adjusting the asset return term, q_t^i, so that nominal house prices (or stock prices) increased only with the rate of CPI inflation since 1971 (i.e., we keep prices constant in real terms). Note that this is an accounting exercise, not a general equilibrium analysis. Our aim here is to illustrate the potential of asset prices to shift the wealth distribution.

Table 6 shows the measured change in the wealth shares of the three groups relative to 1971 as well as the counterfactual change under two scenarios. In the first scenario, we keep real house prices constant. In the second scenario, we fix real stock prices at their 1971 level. The table shows counterfactual changes in wealth shares under these two corner assumptions. The key message from table 6 is that the asset price effects were potentially large. From 1971 to 2007, rising house prices slowed down wealth concentration in the hands of the top 10% by 2.9 percentage points. Without higher house prices, the increase in wealth concentration at the top would have been four times higher than what we observe in the data. The house price crash after 2007 largely reversed these effects, but even in 2016, the observed increase in the top 10% wealth share of

TABLE 6
CHANGES IN WEALTH SHARES RELATIVE TO 1971

Wealth Distribution	1989	2007	2016
Bottom 50%:			
Observed change	−.3	−.8	−2.2
Constant house prices	−.5	−1.7	−2.8
Constant stock prices	−.2	−.4	−2.0
50%–90%:			
Observed change	2.1	−1.7	−6.1
Constant house prices	1.7	−3.6	−7.6
Constant stock prices	2.6	1.4	−2.9
Top 10%:			
Observed change	−1.8	2.4	8.2
Constant house prices	−1.2	5.3	10.4
Constant stock prices	−2.4	−1.1	4.9

NOTE.—This table shows changes in wealth shares for different wealth groups relative to 1971. "Observed change" shows the observed change in wealth shares between 1971 and the indicated year. "Constant house prices" shows the counterfactual change in wealth shares for constant real house prices at the 1971 level. "Constant stock prices" shows the counterfactual change in wealth shares with constant real stock prices at the 1971 level. Changes in wealth shares are shown in percentage points relative to the 1971 level of the wealth share for each group.

8.1 percentage points was still about one-quarter lower than the counterfactual increase of 10.4 percentage points. Again, it is important to realize the magnitude of these shifts in wealth shares. A difference of 2 percentage points corresponds to over 14% of total annual household income. The last row for each wealth group in table 6 also reports the corresponding counterfactual for constant stock prices. Without the stock market boom, the top 10% wealth share would have been 1.1 percentage points lower in 2007 than in 1971, and even over the whole period, the middle class (50%–90%) would not have lost substantially less ground relative to the top. These counterfactual simulations are suggestive only, but they highlight to what extent the wealth distribution is sensitive to asset price dynamics.

V. Conclusions

This paper makes three contributions to the literature on income and wealth inequality. First, we introduced the SCF+, an extended version of the SCF. The SCF+ as household-level data set covers the financial situation of US households since World War II. The SCF+ complements existing data sets for long-run inequality research that are based on tax and social security records. The SCF+ makes it possible to study the joint distribution of income and wealth over time, as it contains both income and balance sheet information coupled with extensive demographic information. We expect that the SCF+ data will become a valuable resource for future empirical and theoretical research on inequality, household finance, political economy, and beyond.

Second, we exploited the new data to study the trends in income and wealth inequality. Previous research documented a trend toward increasing polarization of income and wealth since the 1970s. The data confirm this finding and underscore that the American middle class was the main loser of increasing income concentration at the top. We also track the racial wealth gap between black and white households over the long run. The picture that the SCF+ paints is one of a persistent income and wealth divide between white and black households. Importantly, we expose divergent trends in the income and wealth distribution for extended periods. Before the crisis in 2008, income growth of households in the lower half of the distribution fell substantially short of wealth growth for these households. However, a large part of these wealth gains has been undone in the crisis. We link these findings to the importance of asset price changes for wealth dynamics.

The third main contribution of the paper is to expose systematic and highly persistent differences in portfolio composition and leverage of households along the wealth distribution. An important consequence of these differences is that asset price changes have first-order effects

on the wealth distribution. They lead to capital gains and losses that induce shifts in the wealth distribution that are unrelated to changes in the income distribution. The magnitude of changes in the wealth distribution induced by this asset price channel can be large. By highlighting the crucial role that portfolio composition, leverage, and asset prices play for the wealth distribution, our paper opens up new avenues for future empirical and theoretical research on the determinants of wealth inequality.

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